MAE 430: Final Project

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Nomenclature

BC	Boundary Condition
BPF	Blade Passing Frequency, Hz
c_p	Specific Heat, J/(kg*K)
d	Diameter, mm
d_h	Hydraulic Diameter, mm
DOE	Design of Experiment
δ	Deformation, mm or m
f	Frequency, Hz
h	Film/Heat Transfer Coefficient of Water, W/m^2°C
H	Height, mm
k	Thermal Conductivity, W/m°C
L	Characteristic length, m or mm
N	Rotational Speed, RPM
Nu	Nusselt's Number
P	Static Pressure, MPa
P'	Harmonic Load, N
Pr	Prandtl's Number
Re	Reynolds Number
s	Arc Length, mm
σ	Stress, MPa
t	Thickness, mm
T	Temperature, °C
mu	Dynamic Viscosity, Pa*s
v	Velocity, m/s
ν	Kinematic Viscosity, m^2/s
z	Number of Impeller Blades
Subscript	
$base$	Base of the Impeller
$blade$	Blade of the Impeller
eq	Equivalent
i	Inner
o	Outer
$mean$	Mean of a set of the given parameter
n	Natural, in reference to natural frequency
r	Rotational
$thru$	Through Flow
tip	At the tip of the blades
w	Water

1. Introduction

While finite element methods are often critical for analysis, they can also be used for design optimization. To exemplify this, we were tasked with designing an impeller like the one shown in Figure 1.

An impeller is a component typically used in a larger system, such as a centrifugal pump. By rotating the impeller at high speeds, we can transfer the work put into the impeller to a kinetic energy, speeding up the fluid flowing through the impeller.[1]

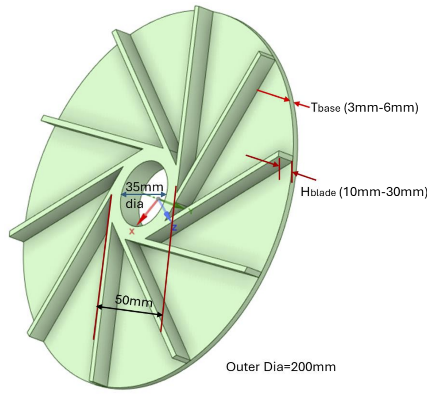


FIGURE 1: A simplified 10-blade impeller geometry with the key dimensions labeled.

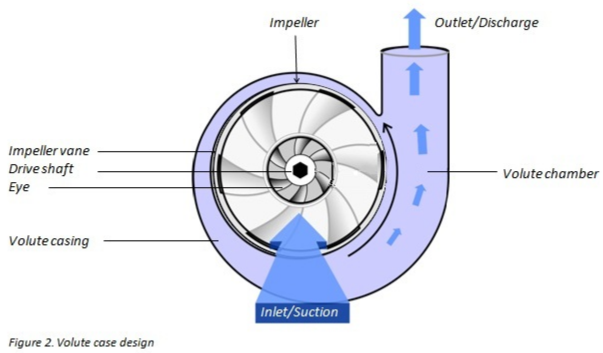


FIGURE 2: A diagram of the flow path in a typical impeller.[1]

As seen in Figure 2, the fluid is typically distributed radially. In order to mount an impeller without causing friction or preventing rotation, impellers are typically attached to a shaft of some sort, like the one seen in Figure 3.

Both the rotation of the motor and the heat of the fluid being pumped can cause a buildup of heat in an impeller. There's a variety of ways that this heat can be alleviated, but one effective method is with forced cooling. Similar to how a heat exchanger transfers heat from one fluid to another, by exposing a side of an impeller to a cool fluid, the impeller can be used to exchange heat from the pumped fluid and other parts of the system to the coolant. The tradeoff for this method is that it requires additional power to pump the coolant and possibly needs extra mechanisms/structures to function.[2]

Other considerations in impeller design include modal analyses to ensure resonant frequencies aren't causing significant distortion. These resonant frequencies are related to the blade-passing frequency (BPF) of an impeller, so by controlling the BPF we can change the resonance and the frequencies they occur at.

1.1. Impeller Design Parameters

For the impeller's design, were provided with the dimensions for some geometry constant throughout the project, shown in Table 1. Since we were asked to optimize the design of a 5 and 10-blade impeller using the ANSYS Workbench Design of Experiment (DOE), we were also given an allowable range for the necessary design parameters not given in Table 1, which are shown in Table 2.

Finally, we are asked to analyze the results of our analysis, with an emphasis placed on harmonic response, separation margins, and overall sensitivities to the geometric differences between DOE impellers.

1.2. Additional Parameters

In addition to the provided design parameters, there were also some parameters needed to conduct analyses and optimization.

The first of these parameters was some missing thermal properties of the fluid flowing through the impeller, water. In order to consider convective boundary conditions due to a fluid, ANSYS Steady-State Thermal simulations require

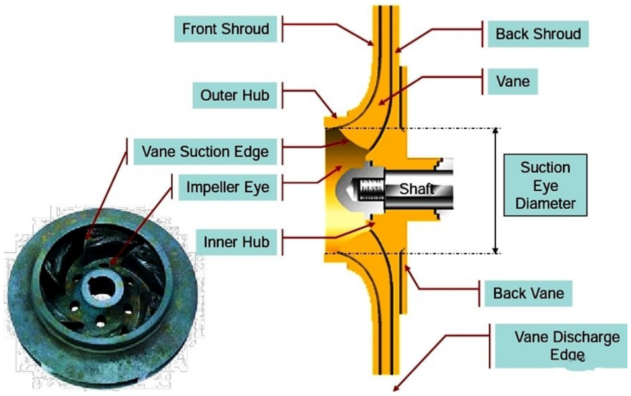


FIGURE 3: An impeller mounted and fed with a hollow shaft. The shaft rotates with the impeller, such that the impeller isn't moving radially or axially, but is still free tangentially.[3]

TABLE 1: Impeller Baseline Design Parameters

Parameter (Units)	Value
d_i (mm)	35
d_o (mm)	200
t_b (mm)	3
Material	Structural Steel
ρ_{water} (kg/m^3)	1000

a fluid's heat coefficient and ambient temperature.

Of the two, the film coefficient was by far the most difficult value. This is due to major variations in water's film coefficient, as the value depends on factors like temperature, pressure, velocity, and surface area. Since all of these factors vary depending on the fluids location in the impeller, we'll have to use a representative value.

We can get an estimate for the film coefficient using the definition of Nusselt's number, a dimensionless parameter often used to characterize the ratio of convective heat transfer to conductive heat transfer. For water, this value is defined as

$$Nu = \frac{h_w L}{k} \tag{1}$$

Where the film coefficient can be calculated experimentally, or Nusselt's numbers calculated based on existing values can be used.[4] By reorganizing Equation 1, we can determine our film coefficient with

$$h_w = \frac{kNu}{L} \tag{2}$$

While we can not take our own experimental data to determine Nu, we can use the Dittus-Boelter equation for a turbulent flow, which is relatively accurate given the heat disparity between the fluid and surface isn't too large.[5] When cooling a fluid, the Dittus-Boelter equation is defined as:

TABLE 2: Impeller Design Variables

Parameter	Range
H_{blade}	10mm - 30mm
N	4000RPM - 8000RPM
P_{mean}	1.5MPa - 2MPa
t_{base}	3mm - 6mm
T_w	80°C - 140°C

$$Nu = 0.023Re^{0.8}Pr^{0.3} \tag{3}$$

For the remaining thermal properties of water, we'll have to make assumptions and use experimental data, as we're not given themal conductivity or viscosity coefficients.

In addition to the thermal properties, we can also calculate othe velocity of the water at the tip of the blades using

$$v_{tip} = \frac{\pi dN}{60} \tag{4}$$

For the mean rotational velocity of 6,000 RPM, Equation 4 evaluates to 62.832 m/s, which will be used for later assumptions.

For harmonic response analysis, we'll also need to use this tip velocity to calculate the excitation pressure. Equation 5 defines the relationship between the tip velocity and that pressure.

$$P' = C'_p \frac{1}{2} \rho v_{tip}^2 \tag{5}$$

The problem statement assumes C'_p to be 0.05.

Finally, the problem statement also asks for us to keep our separation margin above 0.2. The separation margin is defined as

$$Separation\ Margin = \frac{|f_n - f_{BPF}|}{f_{BPF}} \geq 0.2 \tag{6}$$

This means we need to consider the distance

2. Approach/Methodology

To begin this analysis, I first had to make a variety of assumptions about the impeller's geometry and operating conditions, as well as relevant fluid properties.

- The structural steel material listed has the same material properties as the ANSYS Structural Steel material
- The impeller is supported by a shaft on each side, rotating with it. The shafts' geometries are structured in such a way that the impeller can not slide along either shaft. The shaft and impeller can be thought of as a single rigid object rotating at N, with no loadings between the parts.
- The back of the impeller (the bottom of the impeller's base) is submerged in cold water in forced convection. Since there are too many unknowns about the conditions of this coolant to perform and calculations, we'll assume $h = 1000 \frac{W}{m°C}$ (which is within existing acceptable ranges for water) and the temperature is 15°C.
- The impeller and coolant are oriented in such a way that the coolant water is just touching the impeller enough to transfer heat, but not enough to apply any significant pressure.
- The air's thermal conductivity at 22°C and atmospheric pressure is approximately $0.02602 \frac{W}{m°C}$.
- The water flowing into the impeller will flow through the shaft on the bladed side, as is typical for many basic impellers.[6]
- The water will be discharged radially outward, but the impeller is encased in such a way that the water is not in contact with the side of the impeller's base. Whatever holes/channels allow the discharge of water from the eye are assumed to be present but not considered in the simplified diagram.
- The impeller is cooler than the water.
- The thermal properties of the water at all temperatures and velocities are close enough to the properties at 110°C rotating at 6,000 RPM to be used for all temperatures in the range listed in Table 2. These properties are listed in Table 3.
- Once the water is being circulated in the impeller blades, it is distributing its excitation pressure evenly across the blade surfaces and the base of the impeller on the bladed side.

TABLE 3: Hydraulic Diameter Variables

Parameter	Value
${k_w}^7$	$0.617\frac{W}{m\,^\circ C}$
c_p	$4182\frac{J}{kg\,^\circ C}$
${mu_w}^8$	$2.548 * 10^{-4} Pa * s$
$v_{tip} = \frac{\pi d N}{60}$	$62.832\frac{m}{s}$

- The through flow velocity is equal to the flow coefficient times the tip velocity. The coefficient for a centirfugal impeller is typically between 0.01 and 0.1,[9] but we'll assume a coefficient of 0.01 due to impellers having vastly different geometries from pipes. The lack of streamlined flow like in a pipe implies that through velocity should be significantly slower than a pipe, making the low end of this range the ideal choice. This means $v_{thru} = 0.62832\frac{m}{s}$
- All faces not in contact with the water or exposed back are adiabatic.
- The fluid is evenly distributed in the impeller such that its static pressure is applied evenly to all surfaces the water is in contact with.
- Since we are not given any damping information, we'll assume that there is no additional damping in the system aside from the material
- We will assume the damping ratio of structural steel is 0.0015 based on known ranges.[10]

2.1. Designing the Impeller

With my assumptions made, I then created my coupled thermal-structural-modal simulation in Workbench. I did not create the harmonic simulation yet, as it would not be used in the Design of Experiment (DOE), only for the 3 sets of suggested parameters later. I linked all of the simulations, then opened SpaceClaim by editing the geometry of the thermal simulation.

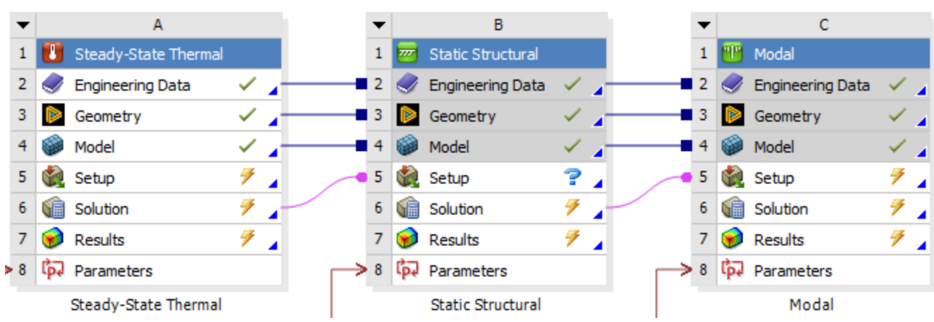


FIGURE 4: The basic coupled simulations needed for a full analysis of the impeller.

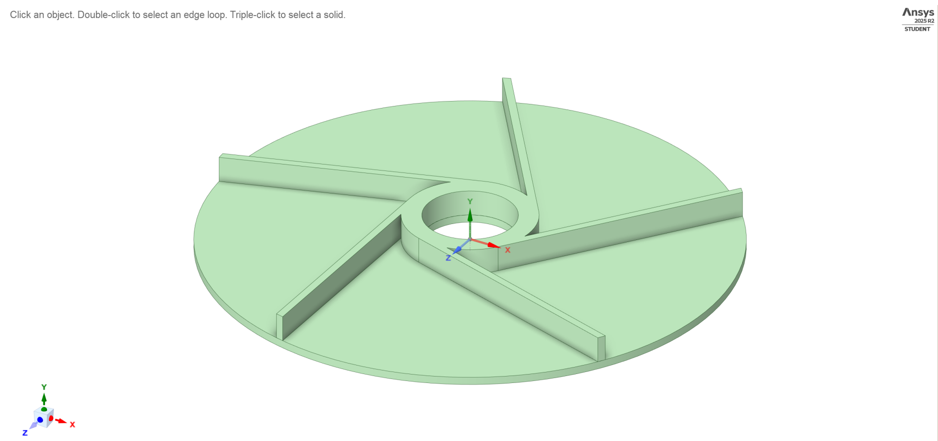
Using the baseline parameters from Table 1, I designed the two variants shown in Figure 16. Since our investigation will span a variety of values for the design variables, I decided to start the design variable parameters at the lowest values in the ranges given by Table 2. Since I designed my impeller in SpaceClaim, it was easy to parameterize dimensions by using the pull tool and ruler. These dimensions were named *DS_base_thickness* and *DS_blade_height*.

I designed the blades first, making them tangential to the 50mm outer diameter tube in the center. I then made the base as a 200mm circle, extruding in the opposite direction from the blades.

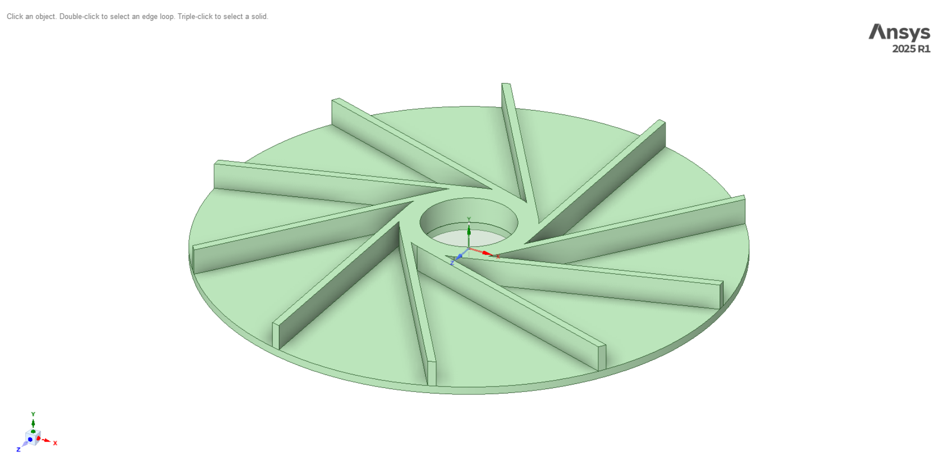
For the sake of brevity, all remaining methodologies discussed after this are the same for both the 5 and 10-blade impellers. As such, the remaining media will only depict the 5-blade impeller.

2.2. Steady-State Thermal Simulation

After saving the geometry, I moved on to setting up the conditions for the steady-state thermal simulation. Since the elements seemed to be big enough that there was only one element along the width of the blades, I changed the element size to 3mm, shown in Figure 6. All other meshing properties were left as the default.



(a) Impeller Geometry: z = 5



(b) Impeller Geometry: z = 10

FIGURE 5: A trimetric view of the initial impeller designs when z = 5 (left) and 10 (right).

For the thermal boundary conditions, I first applied a convection condition to the bottom of the impeller base. This is for the forced cooling listed in the assumptions. I entered, the film coefficient as 1000 and the ambient temperature as 15. Neither of these were parameterized.

Next, I added a convection condition for the surface of the blades, base, and eye in contact with the water being pumped. These surfaces are shown in Figure 7.

To calculate the film coefficient, we can use Equation 2 and the water properties listed in the assumptions. We also need to calculate the characteristic length, which we'll assume to be the hydraulic diameter since we're treating the water as if it's flowing through a pipe.

$$d_h = \frac{4 * Area}{Perimeter} \tag{7}$$

We'll consider the channel's width the arc length at the mean radius between two blades, and the channels height the length from that radius to the end of the blades. Using these dimensions as our channel will give us a generally representative value that leans in the overestimate direction, meaning that inaccuracies will only result in an over-designed impeller

To calculate the mean radius, I used the radii of the hub (the 50mm inner circle) and radius of the base:

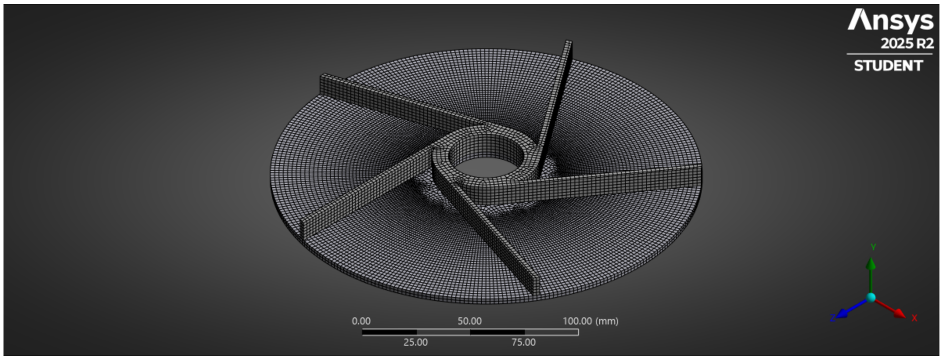


FIGURE 6: The meshed impeller with the default element size used for the mesh. All remaining mesh settings were also default.

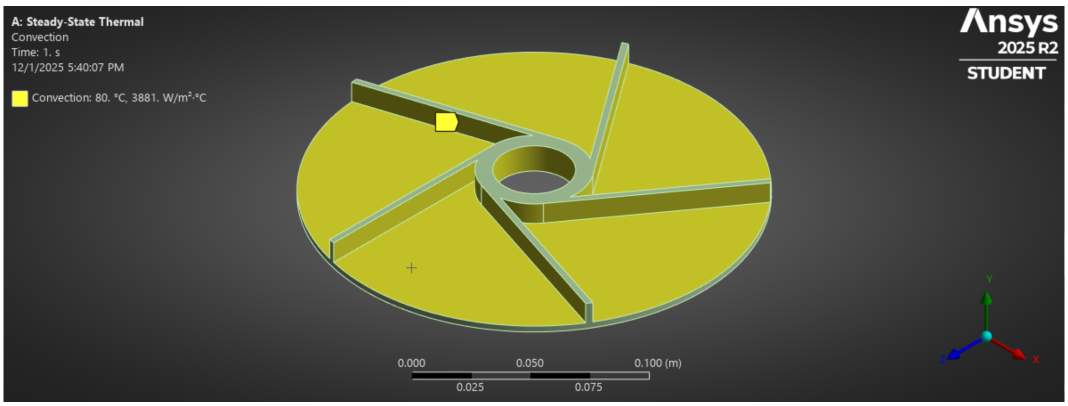


FIGURE 7: The faces selected for the convection boundary condition.

$$
\begin{aligned}
r_{mean} &= \frac{r_o - r_i}{2} \\
&= \frac{100mm + 25mm}{2} \\
&= \frac{125mm}{2} \\
&= 62.5mm
\end{aligned}
$$

The arc length can then be calculated

$$
\begin{aligned}
s_{mean} &= \frac{2\pi r_{mean}}{z} \\
&= \frac{2\pi(62.5mm)}{5} \\
&= \frac{392.699mm}{5} \\
&= 78.540mm
\end{aligned}
$$

We can use the same method to calculate the arc length at the edge of the blades as well.

$$
\begin{aligned}
s_o &= \frac{2\pi r_o}{z} \\
&= \frac{2\pi(100mm)}{5} \\
&= \frac{628.319mm}{5} \\
&= 125.664mm
\end{aligned}
$$

We can treat the area of this slice as approximately a trapezoid and can add the perimeter edges to get the following values:

TABLE 4: Hydraulic Diameter Variables

Parameter	Value ($z = 5$)	Value ($z = 10$
Area	3,828.816mm^2	1,914.408mm^2
Perimeter	279.204mm	177.101mm
D_h	54.853mm	43.239mm

With the hydraulic diameter, I can then calculate the Reynolds number.

$$
\begin{aligned}
Re &= \frac{\rho_w v_{thru} d_h}{\mu_w} \\
&= \frac{(1000\frac{kg}{m^3})(0.62383\frac{m}{s})(0.05485m)}{(2.548 * 10^{-4} Pa * s)} \\
&= \frac{34.465 Pa * s}{2.548 * 10^{-4} Pa * s} \\
&= 1.35264 * 10^5
\end{aligned}
$$

The other value we need to solve the Dittus-Boelter equation is Prandtl's number. This is defined as

$$
\begin{aligned}
Pr &= \frac{\mu c_p}{k} \\
&= \frac{(2.548 * 10^{-4} Pa * s)(4182\frac{J}{kg*K})}{(0.617\frac{W}{m^\circ C})} \\
&= \frac{1.066\frac{W}{m^\circ C}}{0.617\frac{W}{m^\circ C}} \\
&= 1.727
\end{aligned}
$$

Notably, Prandtl's number doesn't include any parameters relating to the number of blades, which explains the identical values in Table 5.

Finally, we have all the values needed to use Equation 3 and calculate the Nusselt number. The components calculated and resulting Nusselt's number are shown in Table 5.

TABLE 5: Dittus-Boelter Equation Variables

Parameter	Value ($z = 5$)	Value ($z = 10$)
Re	$1.35264 * 10^5$	$1.06623 * 10^5$
Pr	1.727	1.727
$Nu = 0.023 Re^{0.8} Pr^{0.3}$	345.035	285.231

The remaining step is to simply plug in all the calculated values. For an impeller with 5 blades:

$$\begin{aligned}
h_w &= \frac{kNu}{D_h} \\
&= \frac{(0.617\frac{W}{m^\circ C})(345.035)}{0.05485m} \\
&= \frac{212.886\frac{W}{m^\circ C}}{0.05485m} \\
&= 3881.006\frac{W}{m^2\,^\circ C}
\end{aligned}$$

I plugged **3881.006** (or **4070.153** for the impeller with 10 blades) into the film coefficient for convection in ANSYS, and entered the ambient temperature as 80°C. I parameterized the ambient temperature.

All other faces were left without any boundary conditions, and were treated as adiabatic. After ensuring both components used the Structural Steel material and adding a temperature distribution to the solver, I ran the simulation. The results are shown in Figure 11a and discussed further in Section 3. I parameterized the maximum temperature value as we're concerned with the maximum stress the impeller is experiencing, which should have a direct relationship with the temperature.

2.3. Static Structural Simulation

With the simulations already linked, the solved thermal loads were then already imported into the static structural simulation. As the mesh was already made, I moved immediately to defining structural boundary conditions.

I first focused on the boundary conditions for powered rotation. I added a cylindrical support to the impeller eyes as the motor's axis would be along the eye. Since I assumed the shafts are rotating with the impeller, which fixes it axially, I fixed all three directions of the cylindrical support. The fix in the tangential direction is allowable as the impeller's rotation is not independent of the shafts, but in sync.

Next, I added a rotational velocity to the impeller's eye to account for centripetal forces. I flipped the direction after selecting the axis geometry such that the impeller was rotating counter-clockwise. I made the magnitude 4000 RPM, and selected the parameterize option.

With the rotational constraints applied, my last task was to apply the fluid loading to the impeller. Based on above assumptions, we can treat the water as an evenly distributed static pressure. Using the pressure boundary condition, I selected all faces in contact with the water, shown in Figure 8. I set the magnitude to 1.5MPa, and selected the parameterize option as the mean static pressure was one of the parameters provided in Table 2.

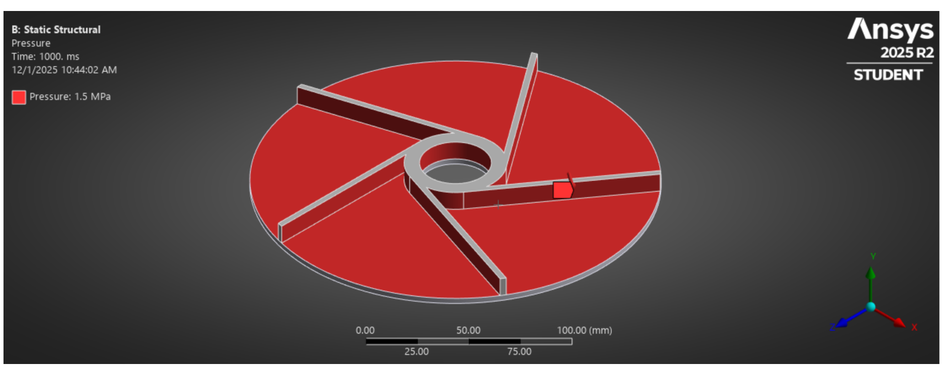


FIGURE 8: The faces the static pressure loading was applied to.

With all the constraints in place to prevent rigid body motion, I added an Equivalent Stress distribution and Total Deformation distribution to the solver and ran the simulation. The results are shown by Figure 12a in Section 3, and I parameterized the maximum value of these results in the solution's *Details* pane.

2.4. Modal Simulation

With my boundary conditions already defined in static structural, the imported pre-stress accounts for all the existing constraints. The one addition between the static-structural and modal simulations was that I enabled damping in the analysis settings, setting the *Constant Structural Damping Coefficient* to 0.0015.

To ensure the first mode found is elastic, I also added a range to the analysis settings between 1 and a sufficiently large maximum (1e9 Hz). As we're only looking for the first elastic node, I limited *Max Modes to Find* to one, which also cuts down on DoE runtime later.

After solving for the first mode, shown in Section 3, I highlighted the mode shape in the graph and selected *Create Mode Shape Results*. I parameterized the frequency and maximum deformation of the first mode shape distribution.

2.5. Design of Experiment Setup

To start my design of experiment, I returned to Workbench and created a standalone Response Surface linked to the design parameters. I entered the ranges in Table 2 into their respective variables in the design of experiment tab. I left the input classifications as continuous. After confirming all input and output parameters were showing, I updated my DoE to run the various data points.

While the 5-blade DoE was running, I moved to another computer to run the DoE for the 10-blade impeller to save time.

After the DoEs were completed, I updated the response surfaces. A snippet of the types of values I found in shown in Figure 9, though the full set of data for both impellers can be found here.

	P1 - DS_blade_height (mm)	P5 - DS_base_thickness (mm)	P6 - RotationalVelocity (rev min^-1)	P7 - Pressure Magnitude (MPa)	P9 - Convection Ambient Temperature (C)	P11 - Equivalent Stress Maximum (Pa)	P12 - Total Deformation Reported Frequency (Hz)	P13 - Total Deformation Maximum (m)
1	20	4.5	6000	2	110	1.5601E+09	1187.283994	3.272716697
2	10	4.5	6000	2	110	2.6545E+09	855.4568998	2.352374449
3	30	4.5	6000	2	110	9.8299E+08	1223.589047	3.104142797
4	20	3	6000	2	110	1.9851E+09	884.3491704	4.18197048
5	20	6	6000	2	110	1.2321E+09	1351.96005	1.473202099
6	20	4.5	4000	2	110	1.5395E+09	1180.78404	3.273941164
7	20	4.5	8000	2	110	1.5890E+09	1196.316442	3.271006786
8	20	4.5	6000	1.5	110	1.1954E+09	1188.998583	3.271994126
9	20	4.5	6000	2.5	110	1.9248E+09	1185.553696	3.273408903
10	20	4.5	6000	2	80	1.5387E+09	1189.057213	3.272412812
11	20	4.5	6000	2	140	1.5815E+09	1185.505999	3.273020342
12	17.16660579	4.074990869	5433.321159	1.85833029	118.5001826	1.9318E+09	1078.134433	3.435470051
13	22.83339421	4.074990869	5433.321159	1.85833029	101.4998174	1.3427E+09	1124.431839	3.495177934
14	17.16660579	4.925009131	5433.321159	1.85833029	101.4998174	1.6253E+09	1184.945123	1.673934078
15	22.83339421	4.925009131	5433.321159	1.85833029	118.5001826	1.1948E+09	1286.856793	3.009245158

FIGURE 9: The first 12 data points of the 5-blade impeller design of experiment.

#	P2 - BaseThickness (mm)	P3 - BladeHeight (mm)	P6 - Rotational Velocity Magnitude (rev min^-1)	P11 - Convection Ambient Temperature (C)	P12 - Pressure Magnitude (MPa)	P9 - Equivalent Stress Maximum (MPa)	P14 - Total Deformation Maximum (m)	P15 - Total Deforr	P16 - Total Deform
1	4.388888889	20	4814.814815	121.1111111	1.694444444	423.7394415	0.0001336813053	1632.334141	1.032914605
2	4.944444444	29.62962963	5111.111111	107.7777778	1.509259259	384.9806297	0.0001244616077	2020.575045	1.41921152
3	3.055555556	11.85185185	6740.740741	123.3333333	1.583333333	418.7828947	0.0001405017486	1122.501876	1.763317902
4	4.611111111	22.22222222	6592.592593	132.2222222	1.712962963	467.7866603	0.000156983845	1747.520489	1.191510614
5	3.277777778	21.48148148	5407.407407	81.11111111	1.601851852	254.053422	0.00008768770271	1712.932447	0.5661658127
6	5.833333333	17.03703704	7481.481481	127.7777778	1.675925926	466.7735648	0.0001515952315	1500.719339	0.3280799477
7	4.5	25.92592593	4074.074074	94.44444444	1.972222222	319.726034	0.00009790410518	1908.706308	1.054360009
8	5.722222222	16.2962963	5851.851852	85.55555556	1.953703704	283.1703705	0.0000907537988	1457.396262	1.269080837
9	3.833333333	11.11111111	6000	134.4444444	1.731481481	469.5630599	0.0001479146828	1082.126016	1.61230092
10	3.611111111	26.66666667	7925.925926	103.3333333	1.898148148	365.4998268	0.000151056561	1933.638289	1.600200743
11	4.055555556	27.40740741	4962.962963	130	1.768518519	487.1848243	0.0001456476615	1949.268696	1.489714795
12	3.5	15.55555556	7185.185185	92.22222222	1.935185185	305.5312015	0.0001084631883	1365.79821	0.6941529775
13	4.277777778	28.88888889	6444.444444	98.88888889	1.62037037	339.157033	0.0001305034022	2009.76033	1.35592142
14	5.055555556	18.51851852	7333.333333	101.1111111	1.861111111	341.6932434	0.000119810664	1562.087316	0.9734358127
15	3.944444444	14.81481481	5703.703704	118.8888889	1.990740741	423.7495156	0.0001317448847	1316.694076	1.535177553
16	5.5	20.74074074	7629.62963	96.66666667	1.546296296	327.0826725	0.0001182304115	1678.193563	0.8109726011
17	5.388888889	13.33333333	6148.148148	110	1.564814815	373.9337854	0.0001207604046	1287.57353	0.4407748134
18	4.833333333	25.18518519	4370.37037	116.6666667	1.805555556	417.4999951	0.0001263312078	1872.842651	1.011968846
19	3.388888889	19.25925926	7777.777778	112.2222222	1.87962963	380.5004739	0.0001405743493	1591.880603	0.2512797143
20	3.166666667	23.7037037	4518.518519	125.5555556	1.657407407	453.3225947	0.0001365613537	1820.192501	1.509744332

FIGURE 10: The first 20 data points of the 10-blade impeller design of experiment.

Looking at the response surfaces, I noticed a variety of trends, which are discussed in Section 3.1.

2.6. Harmonic Response Simulation

To observe how the optimization of different aspects affected the harmonic loading, I ran harmonic response simulations for the values shown in Table 6. Since the 10-blade impeller had superior performance, I just used the 10-blade impeller's 3 optimal designs for program time-efficiency. These values were determined as local optimums by using the ANSYS Response Surface Optimization solver linked to the Response Surface. The optimizer used the *Screening* method at 500 samples. It targeted a minimum stress, minimum resonant frequency, and a minimum deflection at that resonant frequency. The last of these is important because the lowest deflection at the resonant frequency means that the peak at that frequency will be lower, resulting in lower deflections at other frequencies nearby. The minimum resonant frequency choice was a bit arbitrary, as I saw in my response surfaces that most of my natural frequencies were on the order of 10^3, and from experimenting I saw how my blade passing frequency was typically above at least a thousand, increasing significantly as the rotational velocity increases.

TABLE 6: Impeller Harmonic Simulation Parameters

Design	H_{blade} (mm)	t_{base} (mm)	T_w (°C)	N (RPM)	P_{mean} (MPa)	σ_{eq} (MPa)	f_n (Hz)	δ_n (m)	BPF (Hz)	Sep. Margin
1	10	3	80	4000	1.5	234.49	846.73	1.2322	666.667	0.27009
2	10.801	3.483	83.9	7954.6	1.7571	251.34	1079.9	0.83809	1325.767	0.18545
3	10.762	5.403	80.828	6198.8	1.5838	242.84	1142.9	0.70444	1033.1333	0.10625

Notably, designs 2 and 3 are both outside of the separation margin guideline defined in Section 1.2. I'll still perform a harmonic analysis on the two, but Design 1 will be chosen as it is the only one meeting the standard.

To use the design values in a harmonic simulation, I first duplicated my three linked simulations used in the DoE. I de-parameterized the geometry, though I left the values the same for Design 1 as my initial values matched these values. After repeating this process for my other two designs, I would change the parameters to match the values in Table 6. As I refreshed and updated the thermal simulation, I removed the parameterized temperature and left it at 80°C. I did the same for the rotational velocity, leaving its original speed. I de-parameterized the outputs for all three of the simulations before linking a harmonic analysis simulation to the end of this duplicate. I enabled *Large Deflection* in the analysis settings of the Static Structural simulation after some solver warnings.

One issue I had to go back and change was in the modal simulation. As the solver was *Program Controlled*, I couldn't run the harmonic simulation as ANSYS was running *Full Damped*. I manually changed this to *Reduced Damped* and increased the *Max Modes to Find* to 7 to ensure sufficient accuracy with the new solver type.

I then moved on to the harmonic response boundary conditions- specifically the excitation pressure. At 4000 RPM, I used Equation 4 to calculate a tip velocity of 41.888 m/s. Using Equation 5 with this value, the density from Table 1, and the assumed C_p' from the problem statement, I calculated the excitation pressure to be 43.865 kPa. I applied this to the same surfaces I highlighted in Figure 8 except for the eye, where the fluid's movement would be relatively slow before being pulled out radially.

For the analysis settings, I gave a little room around 666.667 Hz to ensure the BPF was in range, so I set the *Range Minimum* to 666.666 Hz and the *Range Maximum* to 666.668 Hz. The solution interval was dropped to one since we're only looking at 1 frequency. To observe the stress and deformations, I placed equivalent stress and total deformation solutions in the solver. The results of this are shown in Section 3.2.

I then repeated this process, replacing the design parameters highlighted in Table 2 with the values in Table 6. I calculated design 2's excitation pressure to be 173.474 kPa, and design 3's excitation pressure to be 105.345 kPa. The stress and deformation distributions of both of these is shown in Section 2.6.

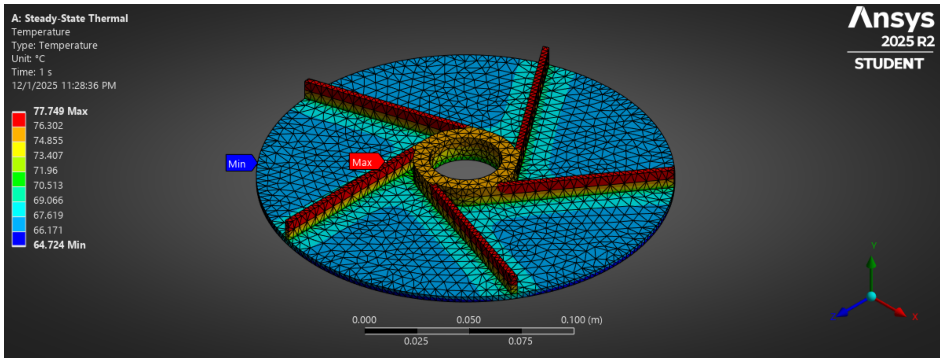
2.7. Errors Located while Repeating Methodology for 10-blade Impeller

With all the simulations done, I repeated my process for the 10-blade impeller. Most of what I did was the same as the above sections, but one notable difference I caught after running all my simulations was that I accidentally left the mesh size of my 10-blade impeller as the default. I left the simulations as in due to time constraints, and was very concerned as the elements weren't that much bigger than the 3mm elements. Still, they were significant enough to note as it'll have some minor effects on the accuracy. When discussing with peers about results, we found that the comparison between my 5-blade and 10-blade results were very similar to their comparisons.

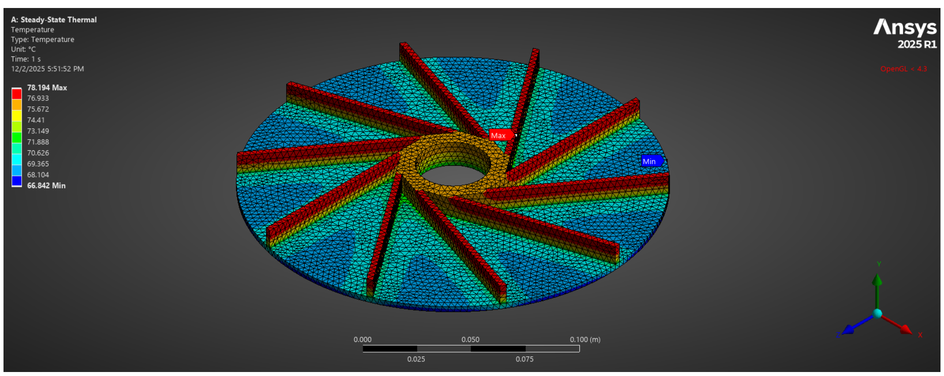
Another note I found when repeating my methodology was a typo in my 5-blade impeller DoE, as I accidentally set the maximum bound for static pressure to 2.5 MPa instead of 2.0 MPa. This did not have a direct effect on the results, and I could still see the data within the initially defined bounds, but it resulted in having some extra data I didn't need in the 5-blade simulation, and I couldn't get more refined data points within the exact range. Still, I was able to read the behavior I needed from the graphs, shown and discussed in Section 3.1.

3. Results & Discussion

Though the problem statement we could expect the impeller blades to reach at least 80°C due to the fluid being accelerated by the impeller, Figure 11 shows how the cooling method we assumed to be used allowed for the fluid to be cooled even more quickly, with both sub-figures showing a blade temperature slightly under the initial temperature of 80° used. The concentrated heat at the blades is due to their distance from the forced cooling being further than the top of the base, which is 10 mm closer to the base in the initial designs shown. Figure 11b boasts a slightly higher temperature at the blade tips, suggesting a relationship between the number of blades and the maximum temperature experienced.



(a) Impeller Geometry: z = 5



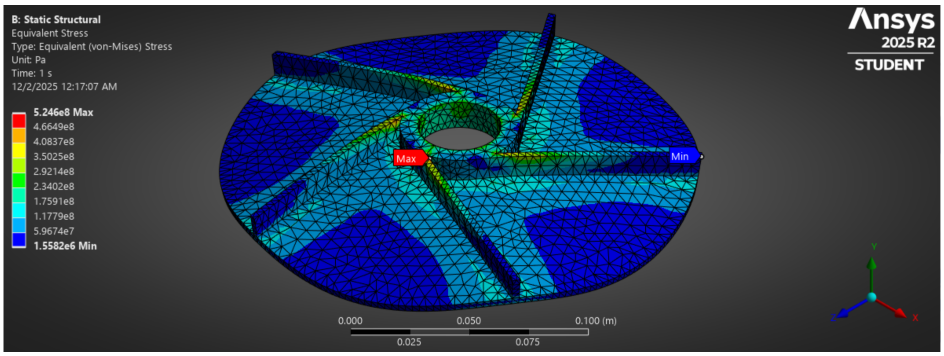
(b) Impeller Geometry: z = 10

FIGURE 11: The temperature distribution of the impellers with 5 blades (left) and 10 blades (right) using the initial parameters defined in Section 2.2.

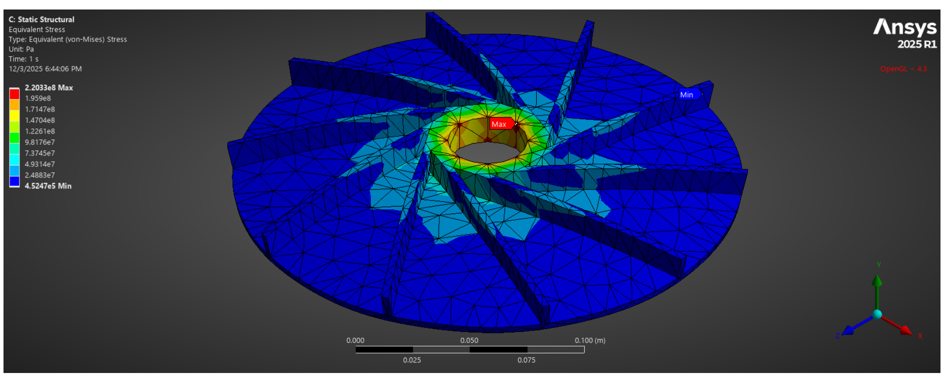
In Figure 12, we can see a much more noticeable difference in the 5-blade and 10-blade impeller distributions. The 10-blade impeller seems to keep the stress more centralized to the eye, whereas the stress of the 5-blade impeller runs along the blades more significantly. This is likely because a similar centripetal force is being generated by each, but the additional blades of the 10-blade impeller are introducing more stiffeners, essentially working against the buckling of each blade. The centripetal force on the 10-blade impeller should actually be larger due to the slightly higher mass, which speaks to how effective the additional blades are at distributing the stress and working against the blades buckling.

Finally, we can see the distortion at the first mode, which speaks to the stiffness of the impeller for each of the blades. While I suspect there's an error with the 5-blade impeller, it could be that the stiffness is just that much less in Figure 13a. This would support my earlier conclusion of the additional blades in the 10-blade impeller acting as stiffeners, reinforcing the base from buckling. That could explain the odd shape in Figure 13a, as ANSYS tends to exaggerate deformations for visual comprehension, and the direction the stiffeners would buckle in is upwards, as seen with Figure 13b.

It is clear, even before performing the design of experiment, that the 10-blade impeller will have superior performance based off the comparative performances of the impellers at the base conditions.

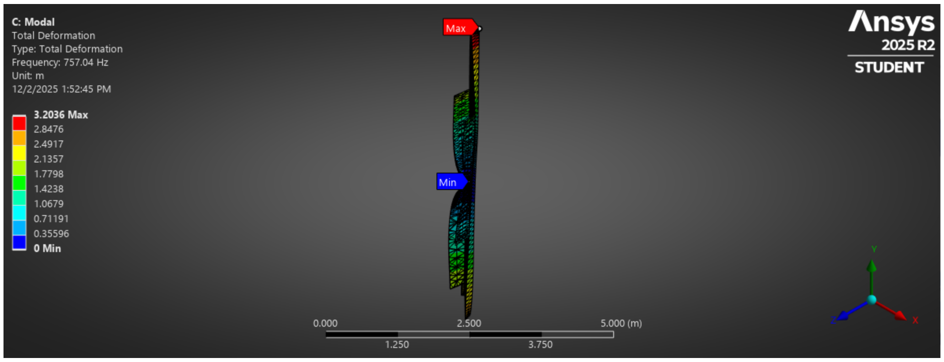


(a) Impeller Geometry: z = 5

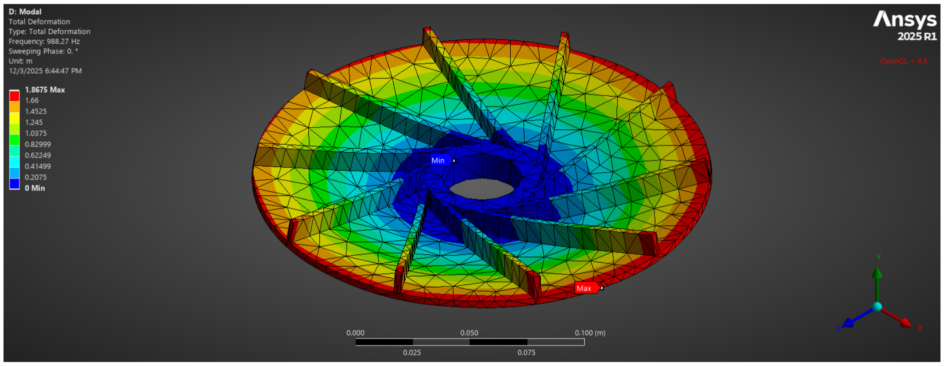


(b) Impeller Geometry: z = 10

Figure 12: The stress distribution of the impellers with 5 blades (left) and 10 blades (right) using the initial parameters defined in Section 2.3.



(a) Impeller Geometry: z = 5

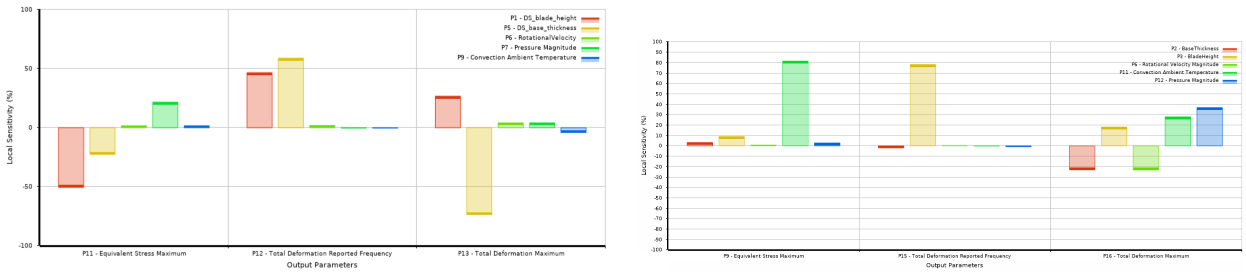


(b) Impeller Geometry: z = 10

FIGURE 13: The displacement distribution of the impellers with 5 blades (left) and 10 blades (right) at their first modal frequencies (757.04 Hz & 937.57 Hz, respectively) using the initial parameters defined in Section 2.4.

3.1. Interpreting DoE Results

To get an initial impression of how strongly each input parameter affects each output parameter, we can look at the Local Sensitivity charts shown in Figure 14. From Figure 14a we can see that the blade thickness and base width of the **5-blade** impeller are the most significant contributors to the stress experienced, the first resonant frequency, and the stiffness at that frequency.



(a) Local Sensitivity Chart of the 5-blade Impeller

(b) Local Sensitivity Chart of the 10-blade Impeller

FIGURE 14: The local sensitivity charts of the impellers. From left to right, the x-axis output parameters are Equivalent Stress Maximum for the Static Structural simulation, Total Deformation Reported Frequency for the first mode in the Modal sim

To interpret how strong these effects are, I looked at the surface of each of the output parameters with the geometry parameters. In Figure 15, there was clearly more weight to the blade height than the base thickness, as the slope along that axis was much steeper. While stress was minimized when base thickness was maximized at 6mm, the slope along that axis was gradual, suggesting our optimized design will focus on increasing the blade height more than increasing the base thickness.

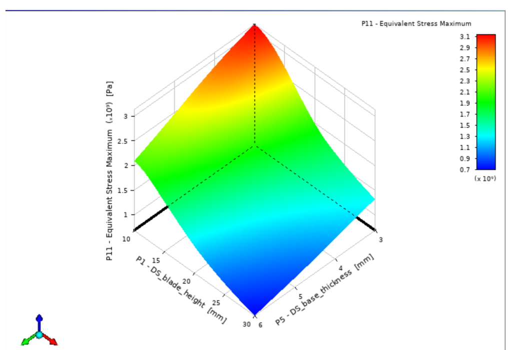


FIGURE 15: The Response Surface of the Equivalent Stress Maximum with respect to geometric design variables.

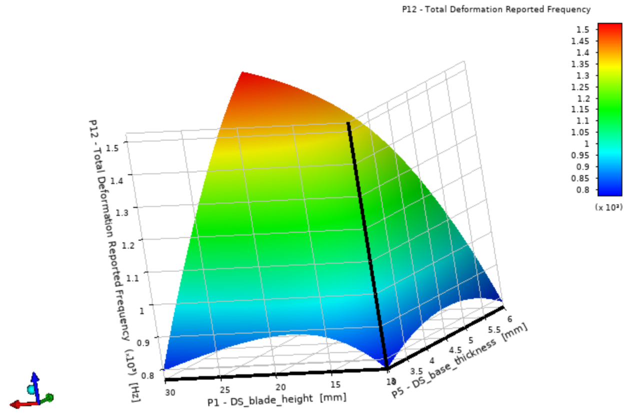
While maximizing the height and thickness seems ideal, we also need to consider the first elastic resonant frequency since we need to ensure it is sufficiently separate from the blade passing frequency (BPF). Figure 16a shows that when both dimensions are maximized, the displacement at the resonant frequency is maximized.

This isn't necessarily a bad thing, as a high resonant frequency may be out of a low operating frequency range. Still, a proximity to the resonant frequency with a high distortion could mean that frequencies nearby have a relatively high distortion as well. Since we're trying to avoid resonance with the BPF, we need to consider the different BPFs at the rotational speeds given in Table 2 first, as done in Section 2.6.

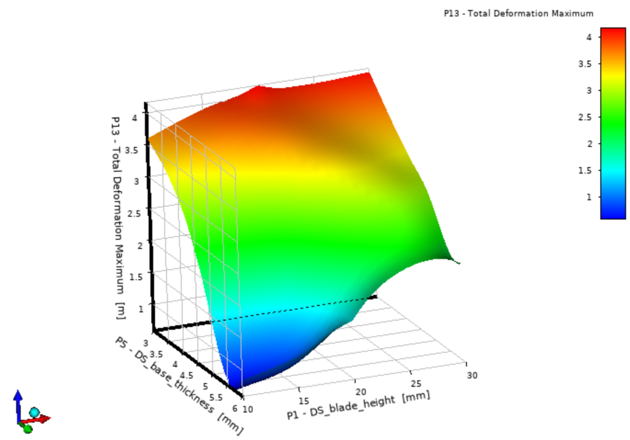
Additionally, we can see how the surface in Figure 16b is much more irregular than the other two response surfaces, suggesting the equivalent stress and resonant frequency rely much more heavily on the geometric parameters than the maximum deformation at that resonant frequency did.

Since the pressure of the fluid being accelerated was the next most notable factor in Figure 14a, I then moved on to look at the temperature's effects. From Figure 17, it is clear the temperature's increase caused an increase in the equivalent stress, meaning that the optimal combination of parameters is likely a balance of the three major parameters instead of using the extremes from any of them.

The other two parameters were shown to have insignificant effects on all three of the output parameters I used. As such, I did not feel the need to compare them to other parameters when discussing behaviors, just that the rotational



(a) Impeller Geometry: z = 5



(b) Impeller Geometry: z = 10

Figure 16: The first elastic resonant frequency (left) and the deformation due to that frequency (right) as a function of the impeller's blade height and base thickness.

velocity and temperature have direct relationships with all three outputs, except for the fluid temperature in the maximum deformation of the first mode, where it is inversely related to the deformation.

Looking at Figure 14b, we see a lot more variation in parameter sensitivity. Unlike the 5-blade impeller, temperature seems to have the largest effect on the stress and total deformation in the static structural simulation. While a bit surprising when compared to how much I expected the pressure to affect the stress, I realize now that with such a low temperature cooling used on the other side of the impeller's base, the increase in temperature would cause a significant increase in temperature difference, a major factor in temperature distribution. For this reason, it makes sense to see a direct correlation between the temperature and stress in Figure 18, whose growth rate looks almost linear.

The sensitivity of the temperature being so strong for the equivalent stress actually explains what I found earlier in Table 6, as the response surface optimize heavily favored low temperatures over all other parameters, since all three designs are near the lower bound of the temperature range.

For the first mode, Figure 14b shows the height having the biggest impact. This is understandable as each blade is acting like a fixed-free column, where the top being free makes the blade much more susceptible to buckling and bending. We see this susceptibility in Figure 19, where the direct relationship seems to have a quadratic growth.

Finally, it was clear that all parameters listed in Table 2 had a noticeable effect on the deformation at the first mode. Figure 14b shows the base thickness and rotational velocity decreasing this deformation as they increase, whereas the blade height, fluid temperature, and static pressure magnitude increasing led to an increase in deformation.

To understand which parameters to prioritizing when designing, we can look at Figures 20, 21, and 22.

We can see in Figure 21 how, as the parameter on each axis increases, the deformation decreases. Despite that, we see a dip on the right side, where base thickness is maximized and the temperature is about in the middle of our range. This is unsurprising, as the base thickness would increase the inertia and subsequent bending moment

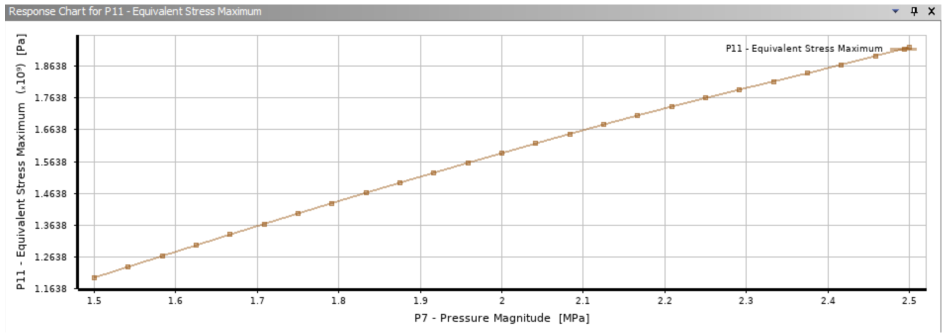


FIGURE 17: The response chart of the Equivalent Stress based on the magnitude of the static pressure of the fluid being accelerated by the 5-blade impeller.

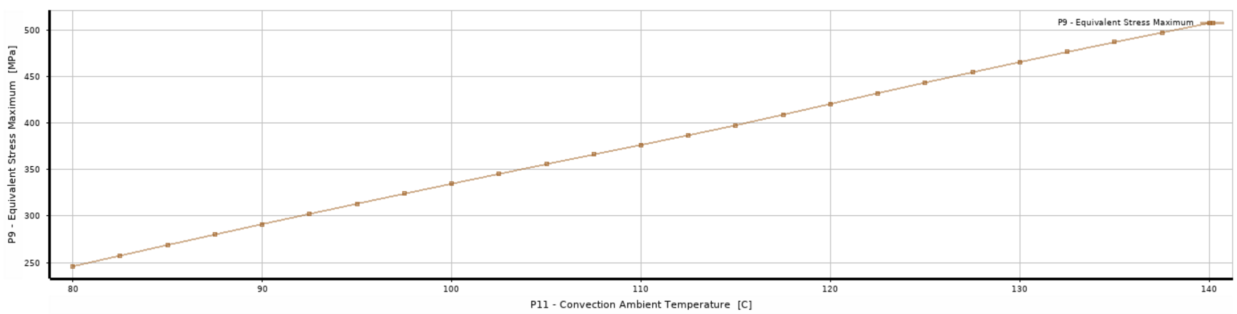


FIGURE 18: The stress response to variations in temperature for the 10-blade propeller.

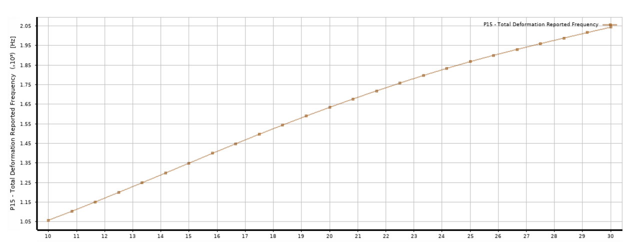


FIGURE 19: The mode 1 frequency response to variations in height for the 10-blade propeller.

needed to bend the impeller. The temperature, on the other hand, depends on the distribution of the temperature throughout the part, which we've seen is not evenly distributed. As the base is stronger than the blades on the impeller, we expect to see the blade failure due to excessive temperature without the entire component bending as severely.

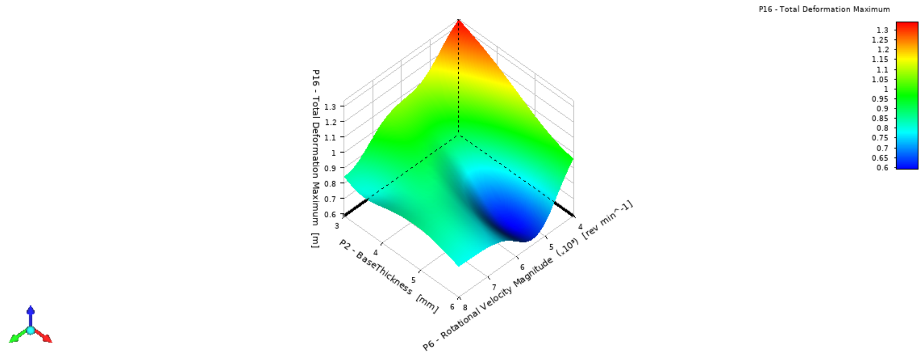


FIGURE 20: The Response Chart of the Total Deformation at the first mode with respect to the base width and rotational speed of the 10-blade impeller.

In Figure 21, we can see how both blade height and temperature increase the deformation at the first node. While the temperature seems to be slightly stronger, the shape is a bit irregular, with a dip in the middle. As we want to prioritize the stress and simply using a rotational frequency to avoid the first node, we can use this dip to compromise on a reasonable maximum deformation when designing without sacrificing other parameters that have more significant effects on other outputs.

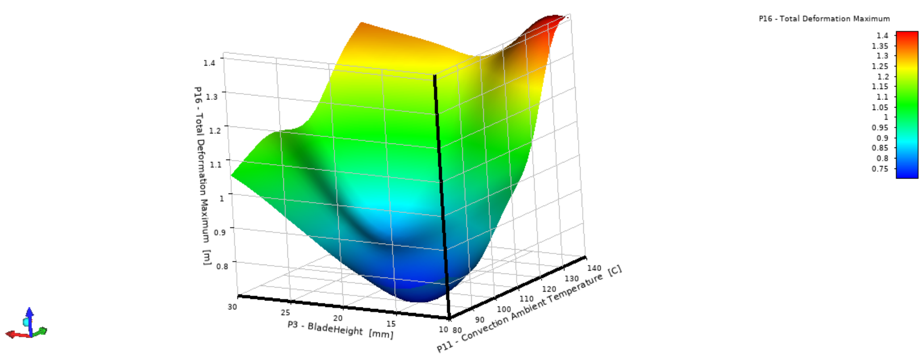


FIGURE 21: The Response Chart of the Total Deformation at the first mode with respect to the blade height and the temperature of the fluid being accelerated by the 10-blade impeller.

Finally, in Figure 22, we see similar behavior to Figures 20 and 21, where the surface has a dip. Surprisingly, this dip is not at the smallest fluid temperature, but closer to the middle of the range. I suspect that at this point, the temperature had some effect that countered the effect of the pressure when at 1.5MPa. For designs where we need to reduce the pressure to improve another parameter, it seems using a temperature in the range of this dip would be preferable. Since temperature has a significant effect on the equivalent stress, though, it seems picking a pressure along the nearby valleys is the best compromise.

While I could pick a value from within these ranges, I chose to use the built-in optimizer to see what types of designs ANSYS found best. For additional details on the response surfaces found from my DoEs, please refer to the two-dimensional response surfaces in the 4. Since the 10-blade impeller shows significantly smaller magnitudes for the equivalent stresses, I only proceeded with the 10-blade impeller, as seen in Section 2.6.

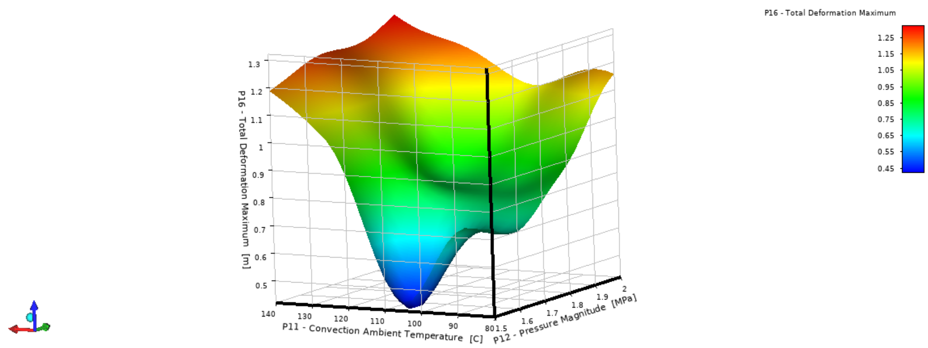


FIGURE 22: The Response Chart of the Total Deformation at the first mode with respect to the base width and rotational speed of the 10-blade impeller.

3.2. Harmonic Excitation Stresses & Amplitudes

The resulting stress and strain at Design 1 is shown in Figures 23 and 24. As this is the only choice within the required separation margin, this is the final design.

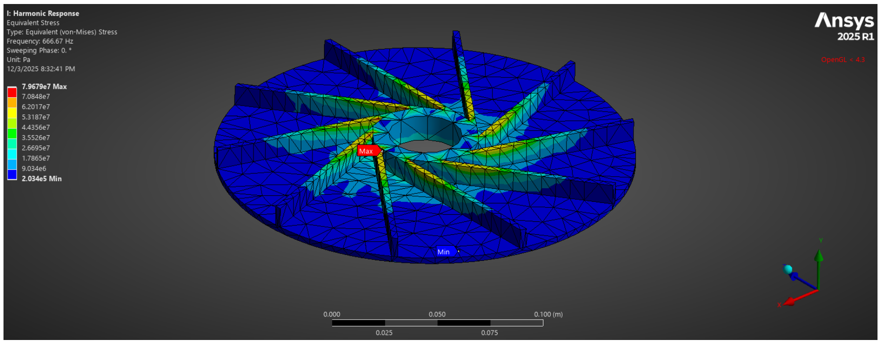


FIGURE 23: The stress distribution in Design 1 from 6 at the BPF of 666.667 Hz.

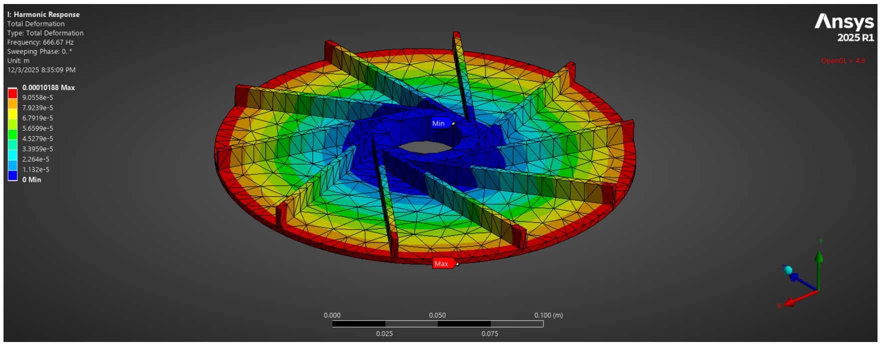


FIGURE 24: The total deformation distribution in Design 1 from 6 at the BPF of 666.667 Hz.

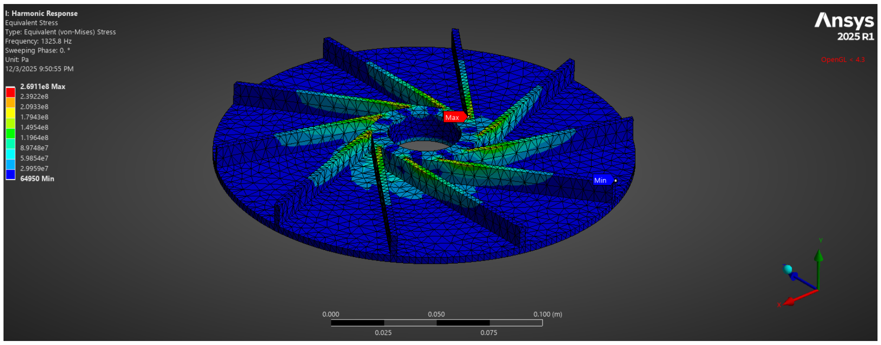


FIGURE 25: The stress distribution in Design 2 from 6 at the BPF of 1325.767 Hz.

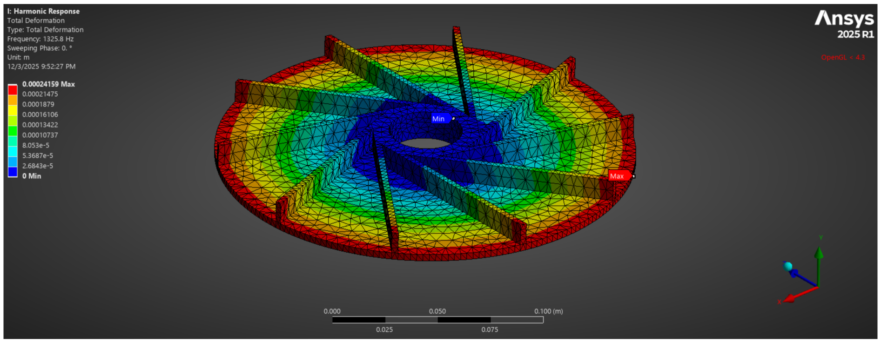


FIGURE 26: The total deformation distribution in Design 2 from 6 at the BPF of 1325.767 Hz.

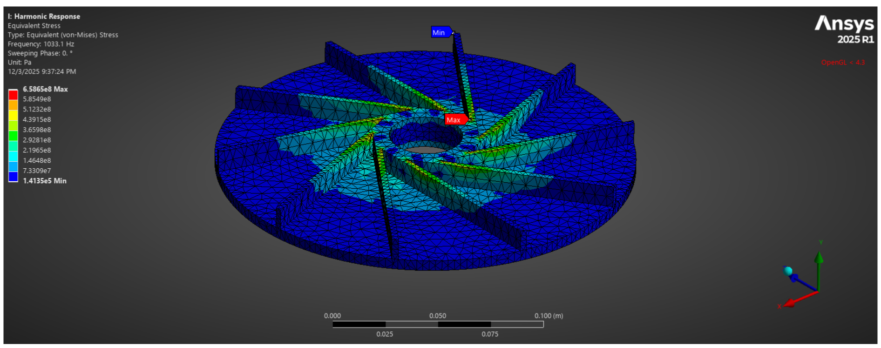


FIGURE 27: The stress distribution in Design 3 from 6 at the BPF of 1033.133 Hz.

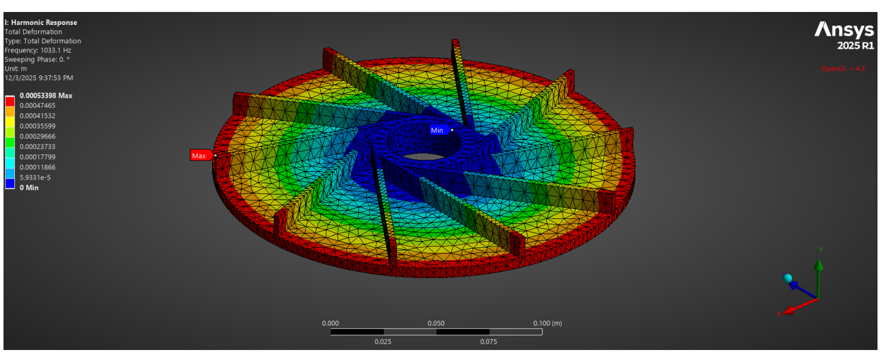


FIGURE 28: The total deformation distribution in Design 3 from 6 at the BPF of 1033.133 Hz.

3.3. Selected Design Distributions

The design that I decided to go with was Design 1 from Table 6. The harmonic stress and deformation at the BPF was already provided in Figure 23 and 26.

The temperature distribution is shown in Figure 29.

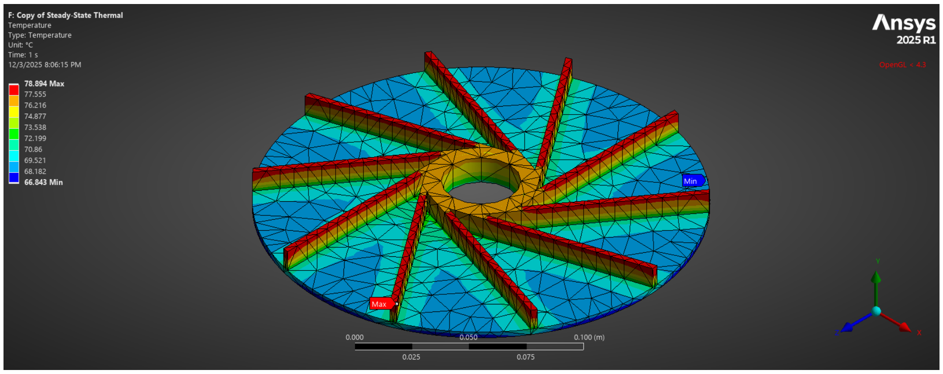


FIGURE 29: The temperature distribution of the optimized impeller, Design 1.

The stress distribution in Figure 30 looks identical to Figure 12b, which is expected as Design 1 matches the initial parameters I used. This design minimized the temperature, which was by far the biggest factor in equivalent stress based on Figure 14b.

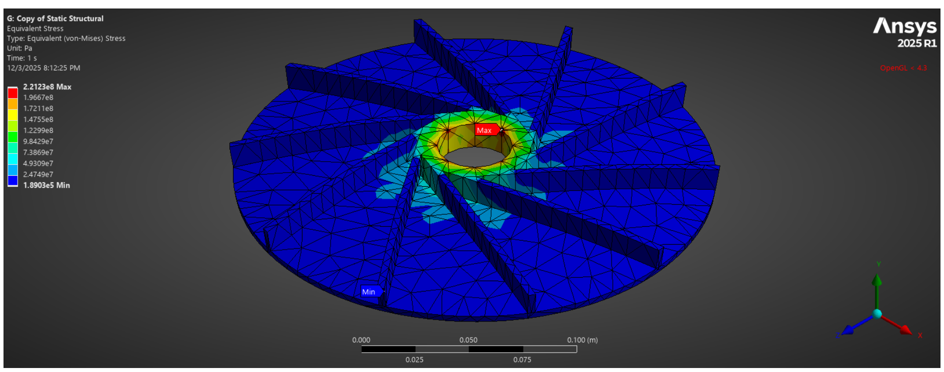


FIGURE 30: The equivalent stress distribution of the optimized impeller, Design 1.

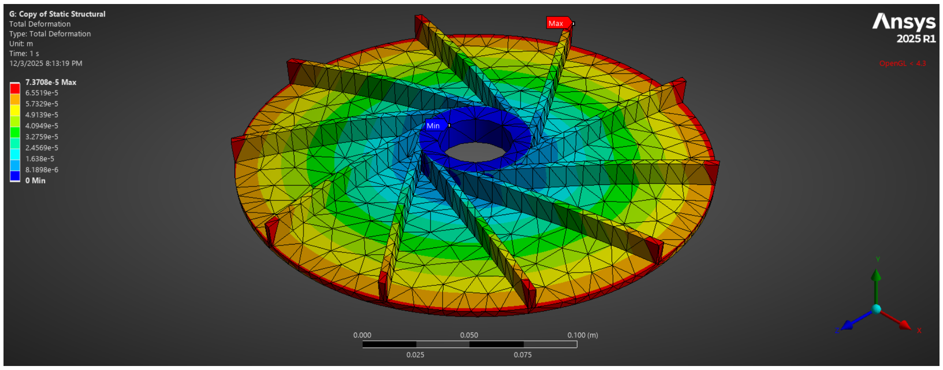


FIGURE 31: The total deformation distribution of the optimized impeller, Design 1.

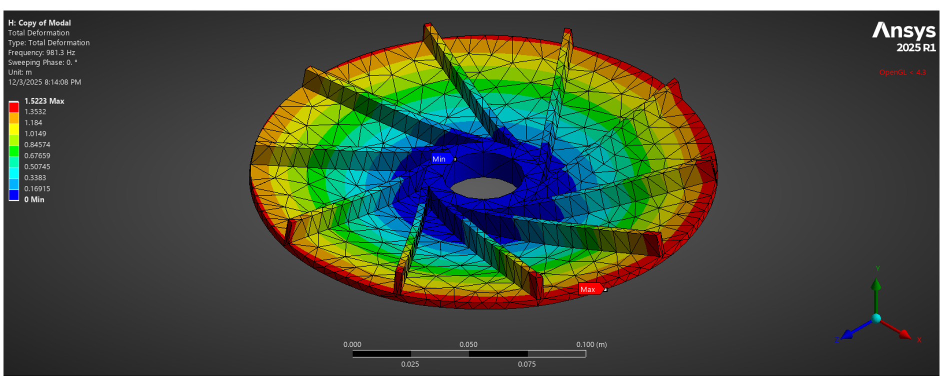


FIGURE 32: The total deformation distribution at the first mode of the optimized impeller, Design 1.

4. Conclusion

It is clear that the ANSYS Design of Experiment and Response Surface Optimizer are powerful tools for optimizing parts and investigating tradeoffs for better performance with certain parameters. From optimizing strength and weight for spacecrafts to determining environmental changes needed for a part to work with a given performance, the vast applications are only as powerful as our own intuition to determine the boundary conditions and loadings. When solving my static structural and harmonic simulations, asymmetrical stress and deformation distributions helped me determine faces I'd missed when setting loading geometry, and the various film coefficients I calculated outside of the typical forced convection range helped me determine errors in my calculations.

Not only does our intuition help us to define the boundary conditions of such simulations, but it also helps us determine where our margins of error may need to be the largest to accommodate for unavoidable accuracy limitations. From recognizing the accuracy drawbacks of faster solving tools to catching software interpretation for parts like the different mesh size I accidentally used for the 10-blade propeller, finite element analysis and design optimization through such methods is only as powerful as we make it.

Appendix

4.1. 10-Blade Impeller 2D Response Surfaces

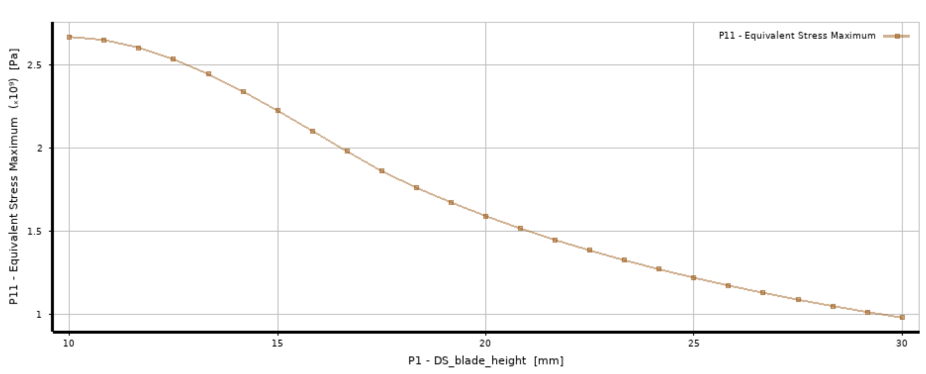


FIGURE 33: The 2D response surface of the maximum equivalent stress due to the blade height of the 5-blade impeller.

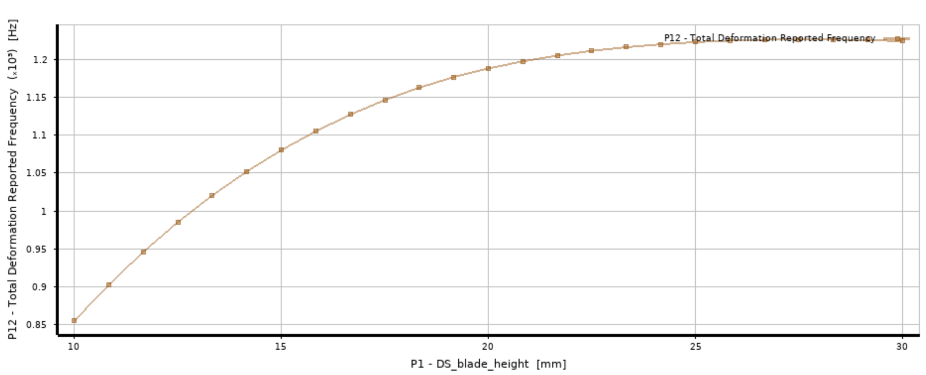


FIGURE 34: The 2D response surface of first mode's frequency due to the blade height of the 5-blade impeller.

4.2. 10-Blade Impeller 2D Response Surfaces

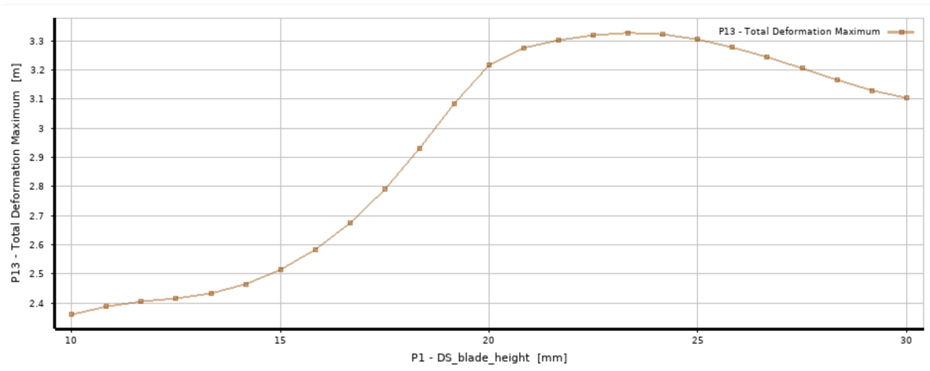


FIGURE 35: The 2D response surface of the maximum total deformation at the first mode due to the blade height of the 5-blade impeller.

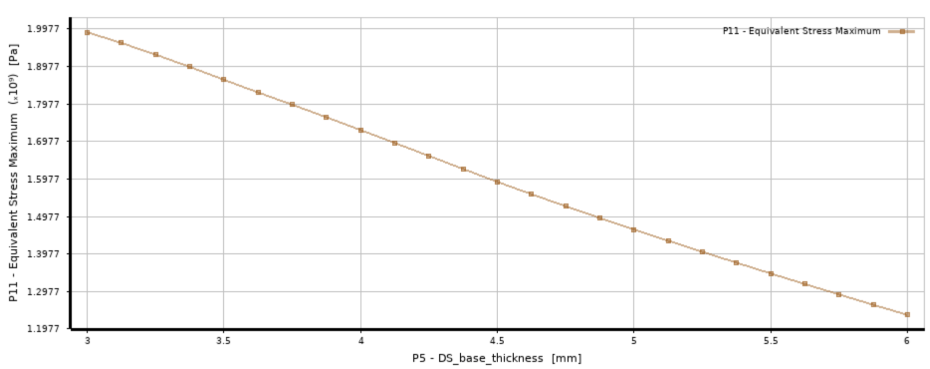


FIGURE 36: The 2D response surface of maximum equivalent stress due to the base thickness of the 5-blade impeller.

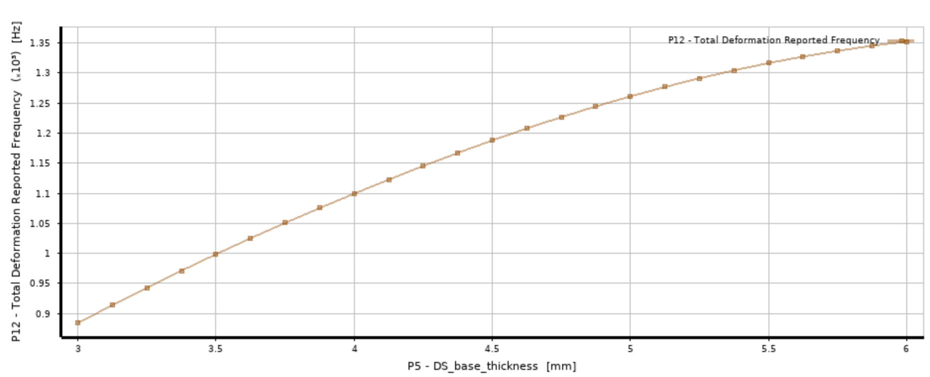


FIGURE 37: The 2D response surface of the first mode frequency due to the base thickness of the 5-blade impeller.

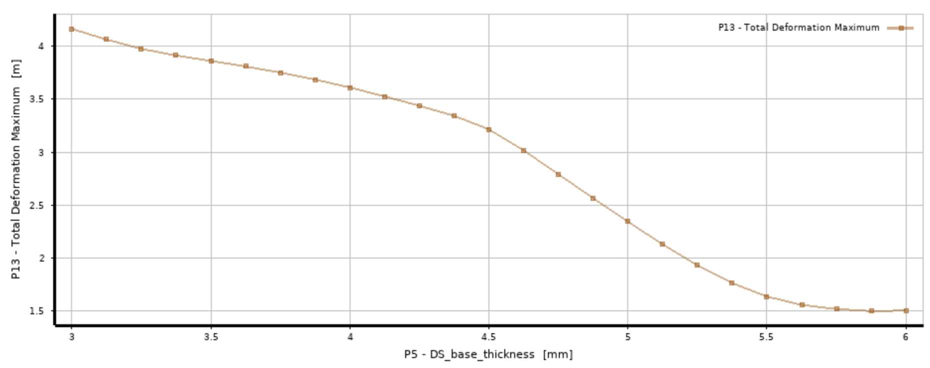


FIGURE 38: The 2D response surface of the maximum total deformation at the first mode due to the base thickness of the 5-blade impeller.

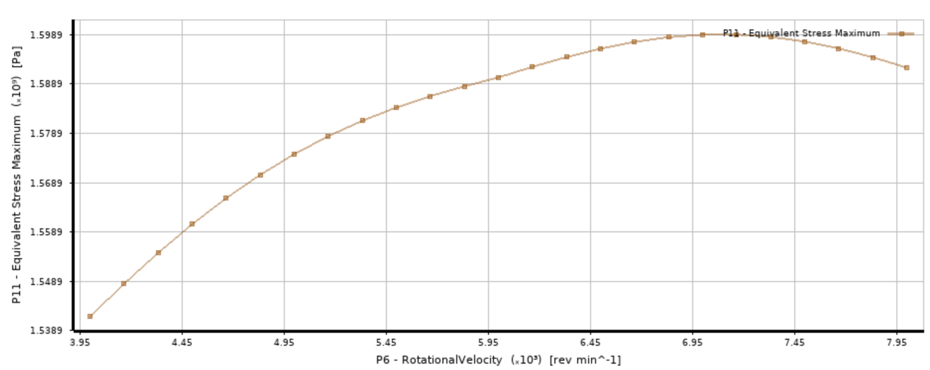


FIGURE 39: The 2D response surface of the maximum equivalent stress due to the operating velocity of the 5-blade impeller.

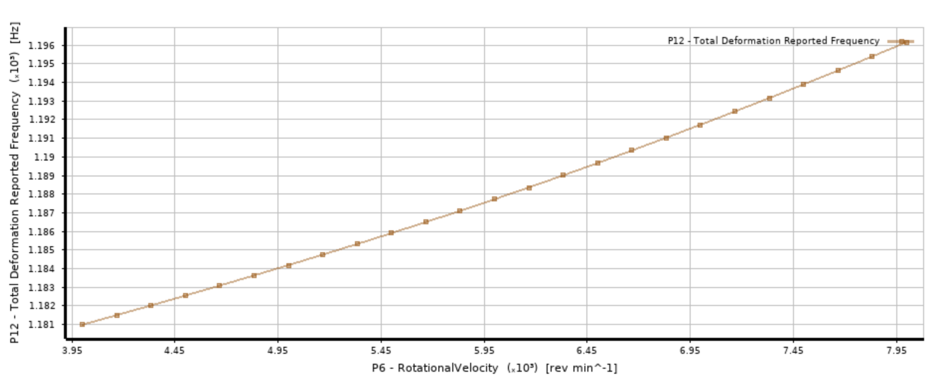


FIGURE 40: The 2D response surface of the first mode frequency due to the operating velocity of the 5-blade impeller.

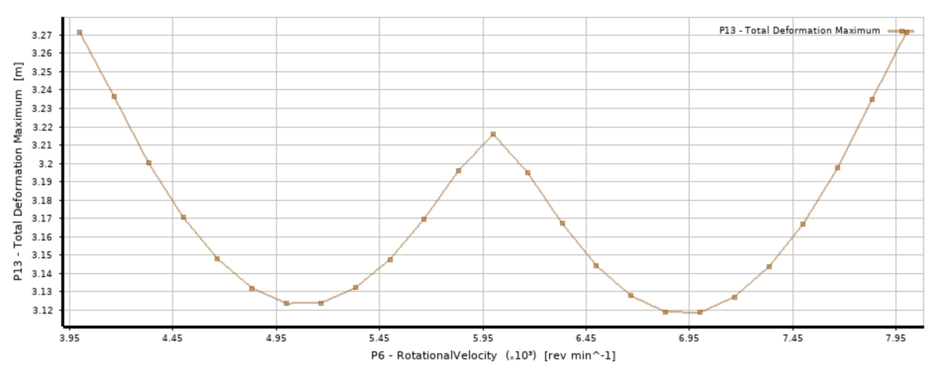


FIGURE 41: The 2D response surface of the maximum total deformation at the first mode due to the operating velocity of the 5-blade impeller.

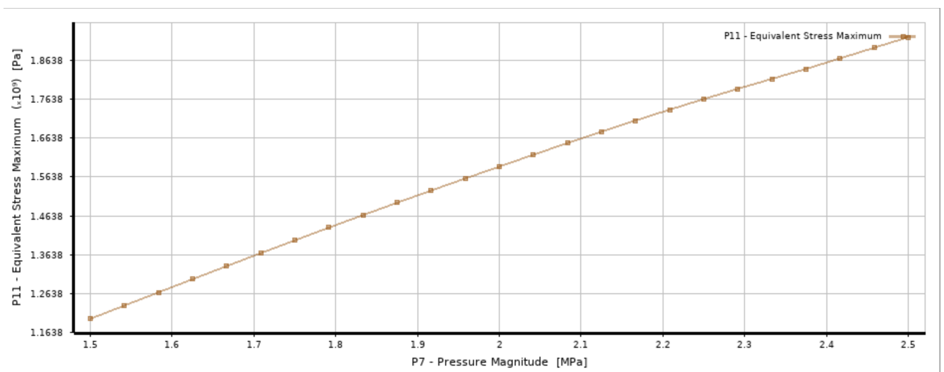


FIGURE 42: The 2D response surface of the maximum equivalent stress due to the static pressure of the fluid being accelerated by the 5-blade impeller.

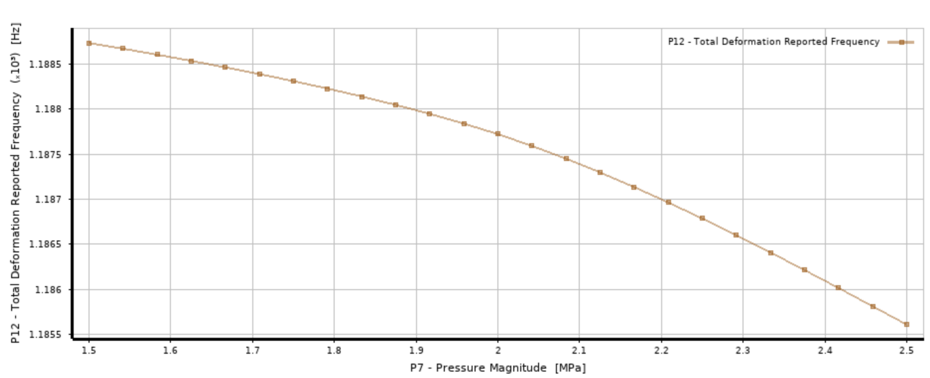


FIGURE 43: The 2D response surface of the first mode frequency due to the static pressure of the fluid being accelerated by the 5-blade impeller.

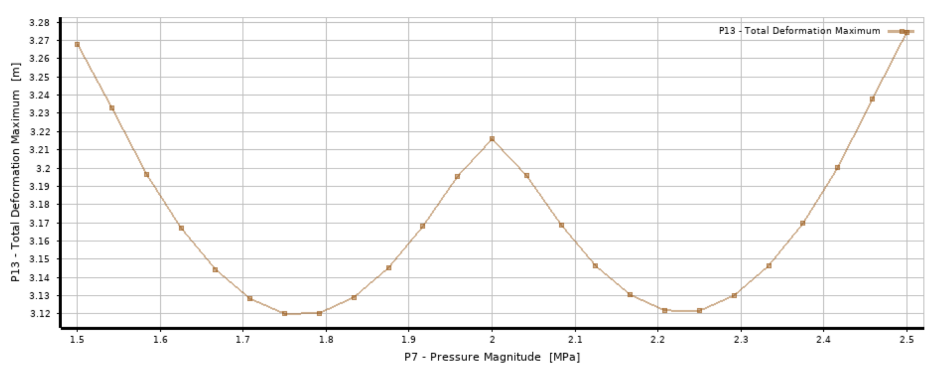


FIGURE 44: The 2D response surface of the maximum total deformation of the first mode due to the static pressure of the fluid being accelerated by the 5-blade impeller.

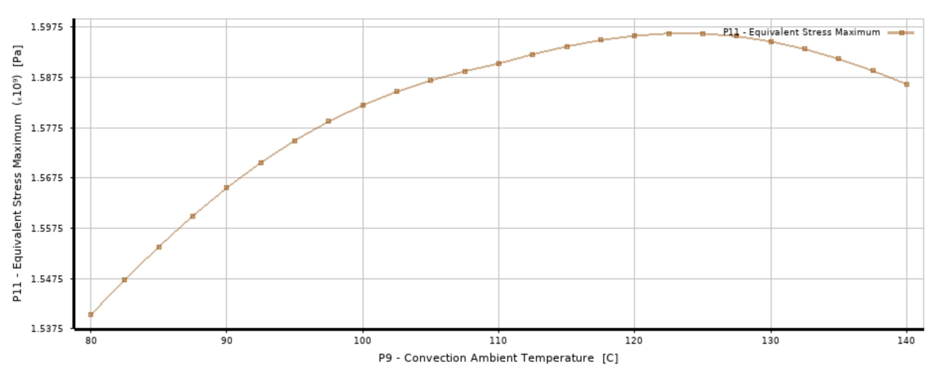


FIGURE 45: The 2D response surface of the maximum equivalent stress due to the temperature of the fluid being accelerated by the 5-blade impeller.

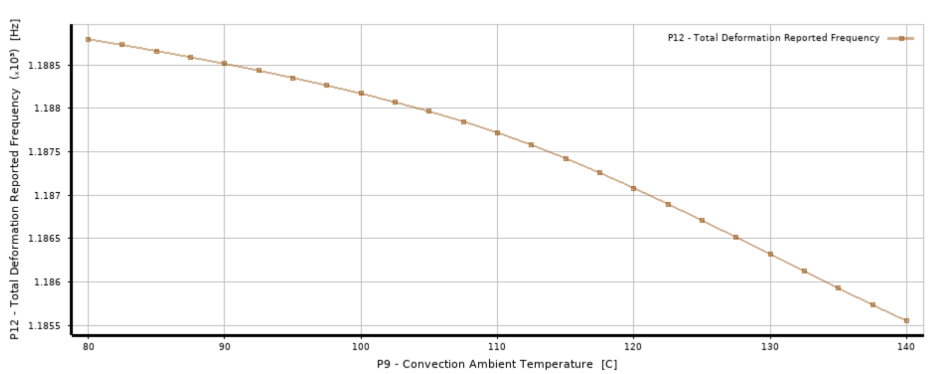


FIGURE 46: The 2D response surface of the first mode frequency due to the temperature of the fluid being accelerated by the 5-blade impeller.

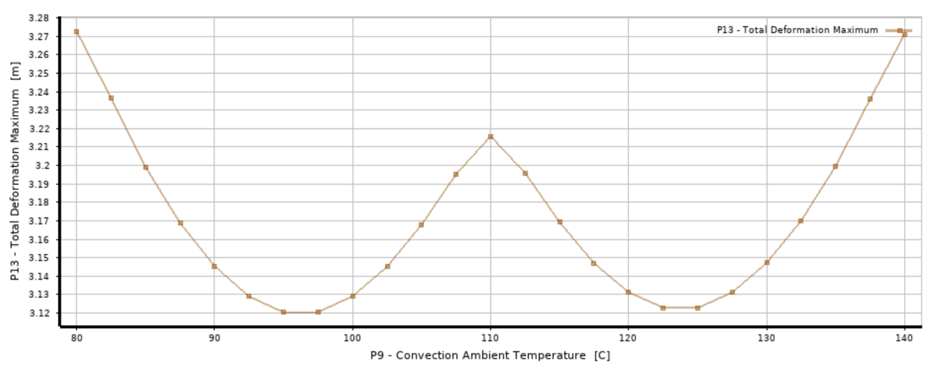


FIGURE 47: The 2D response surface of the maximum total deformation of the first mode due to the temperature of the fluid being accelerated by the 5-blade impeller.

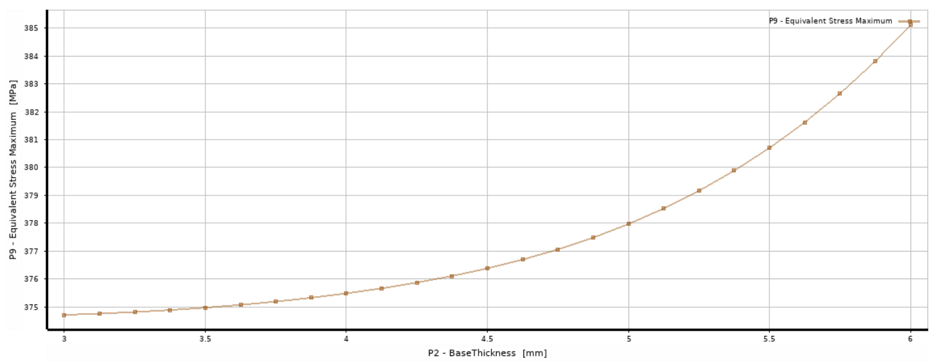


FIGURE 48: The 2D response surface of the maximum equivalent stress due to the base thickness of the 10-blade impeller.

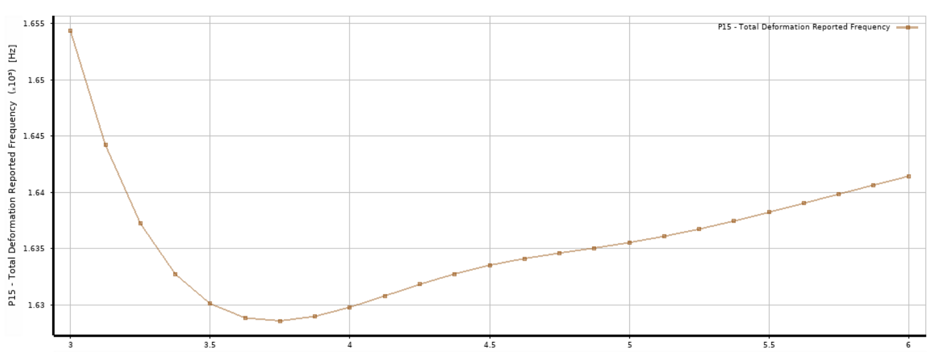


FIGURE 49: The 2D response surface of the first mode frequency due to the base thickness of the 10-blade impeller.

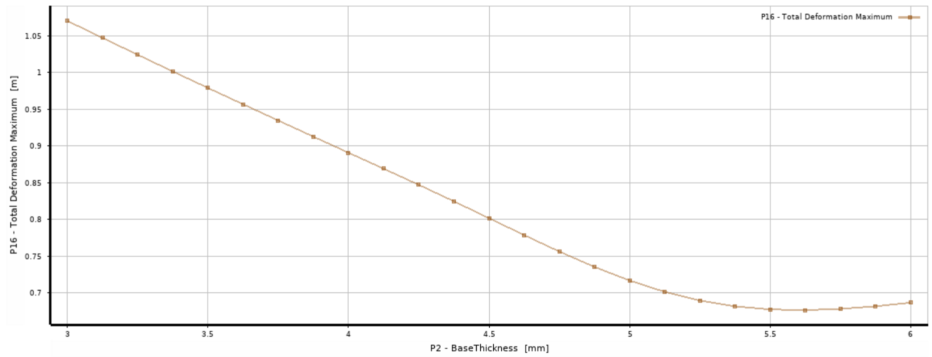


FIGURE 50: The 2D response surface of the maximum total deformation at the first mode due to the base thickness of the 10-blade impeller.

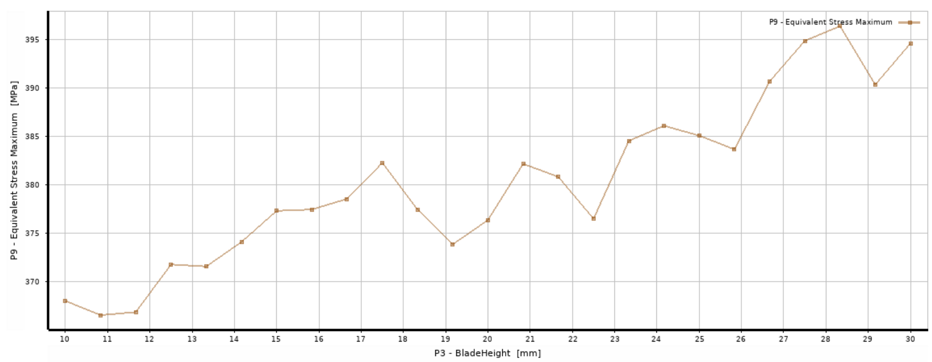


FIGURE 51: The 2D response surface of the maximum equivalent stress due to the blade height of the 10-blade impeller.

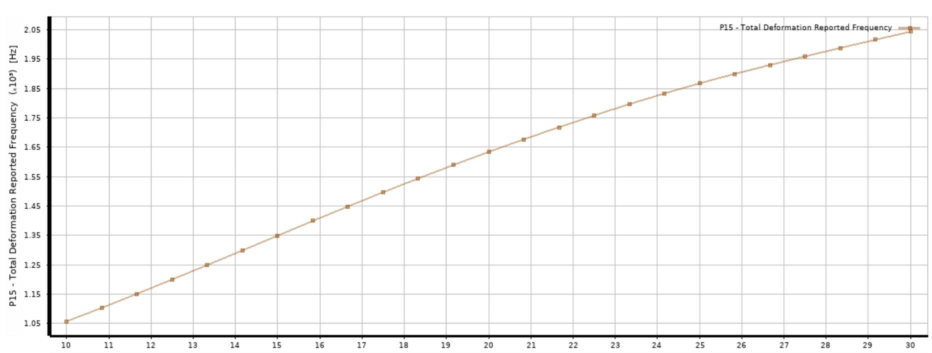


FIGURE 52: The 2D response surface of the first mode frequency due to the blade height of the 10-blade impeller.

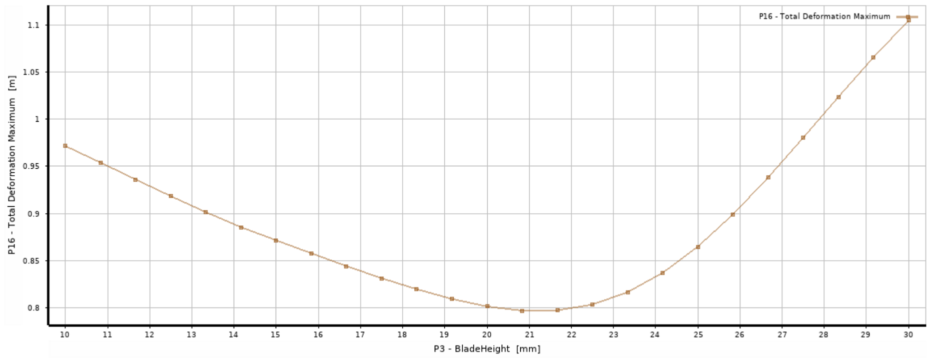


FIGURE 53: The 2D response surface of the maximum total deformation at the first mode due to the blade height of the 10-blade impeller.

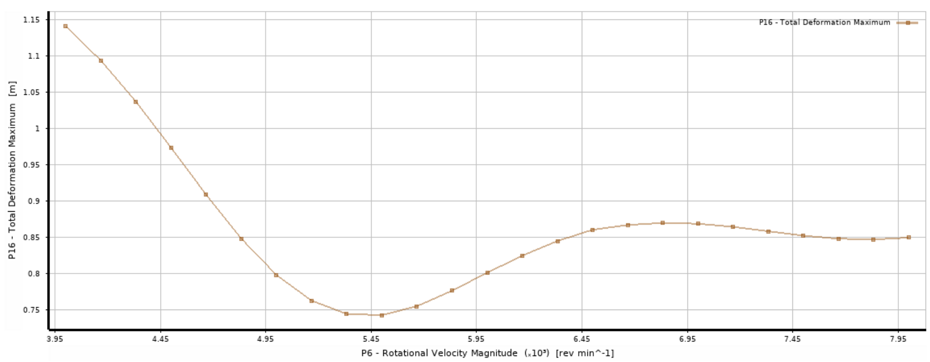


FIGURE 54: The 2D response surface of the maximum equivalent stress due to the operating velocity of the 10-blade impeller.

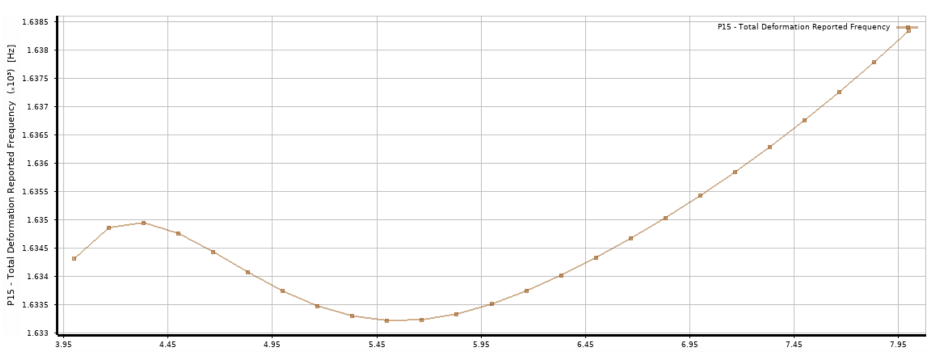


FIGURE 55: The 2D response surface of the first mode frequency due to the operating velocity of the 10-blade impeller.

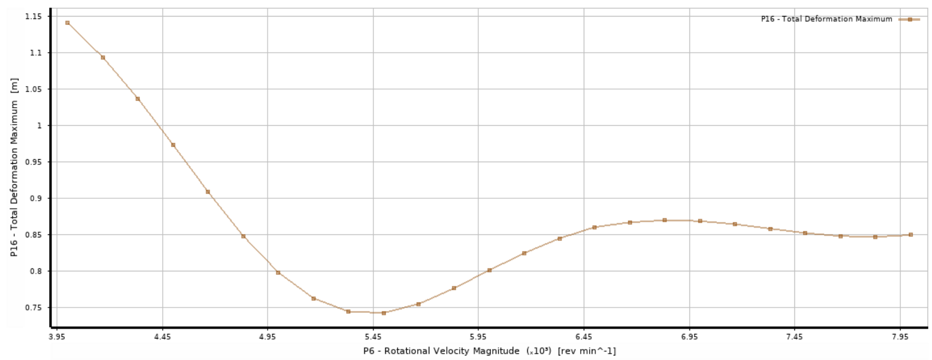


FIGURE 56: The 2D response surface of the maximum total deformation at the first mode due to the operating velocity of the 10-blade impeller.

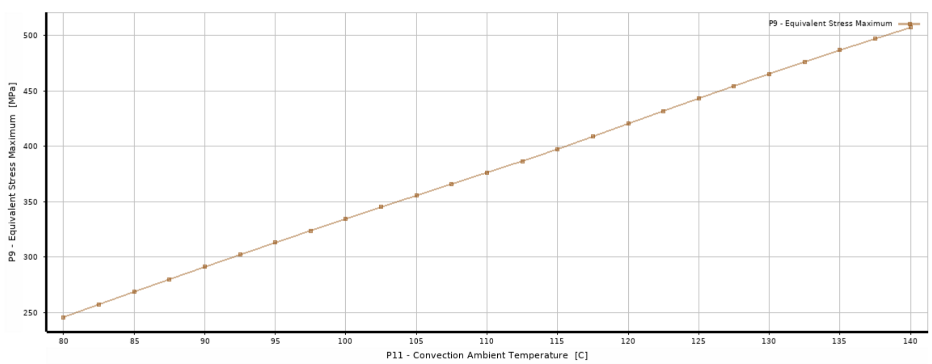


FIGURE 57: The 2D response surface of the maximum equivalent stress due to the temperature of the fluid being accelerated by the 10-blade impeller.

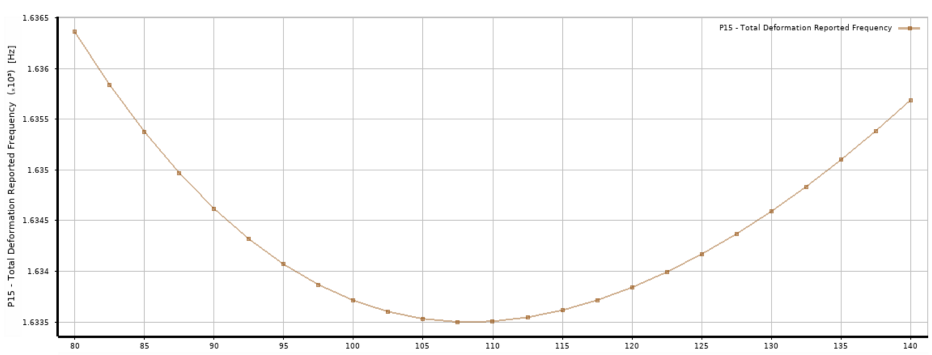


FIGURE 58: The 2D response surface of the first mode frequency due to the temperature of the fluid being accelerated by the 10-blade impeller.

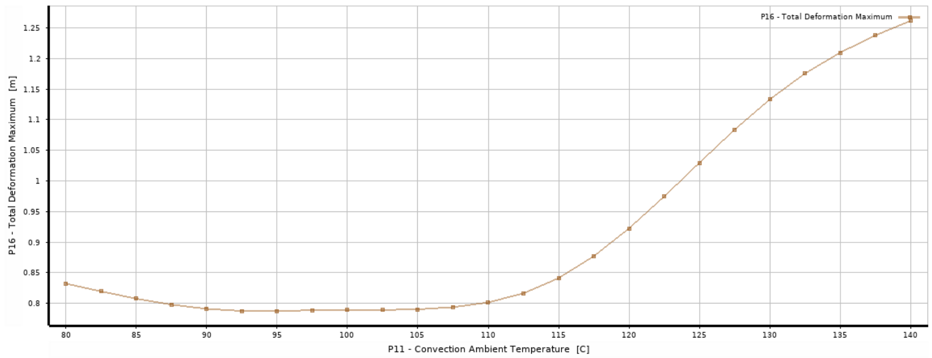


FIGURE 59: The 2D response surface of the maximum total deformation at the first mode frequency due to the temperature of the fluid being accelerated by the 10-blade impeller.

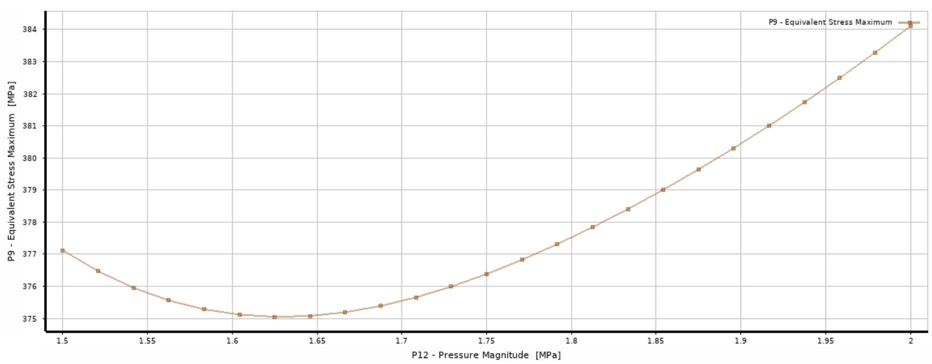


FIGURE 60: The 2D response surface of the maximum equivalent stress due to the static pressure of the fluid being accelerated by the 10-blade impeller.

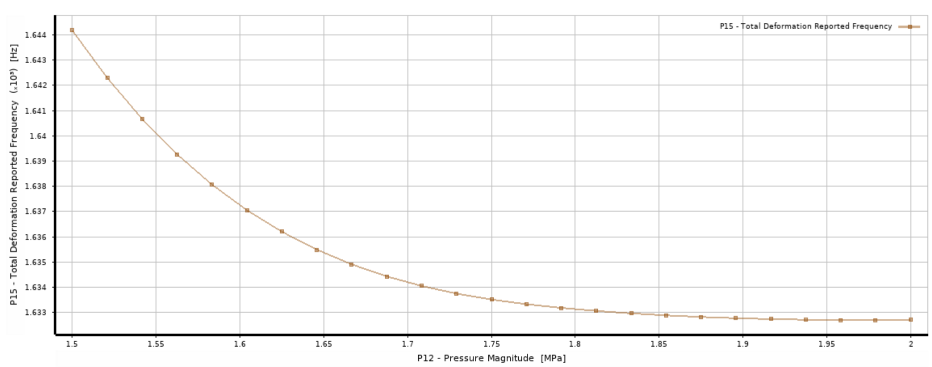


FIGURE 61: The 2D response surface of the first mode frequency due to the static pressure of the fluid being accelerated by the 10-blade impeller.

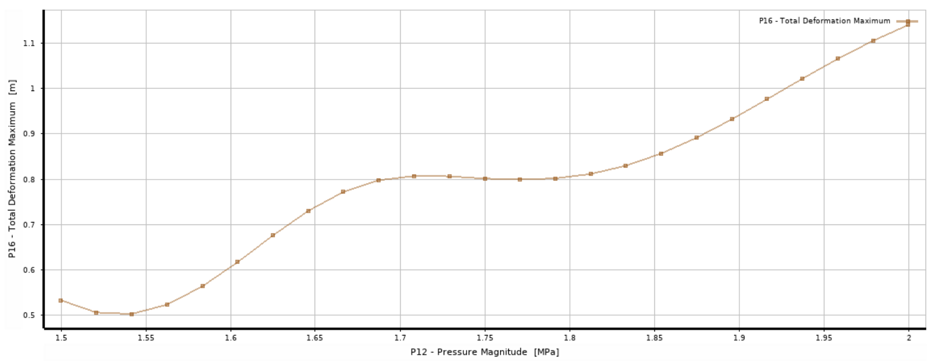


FIGURE 62: The 2D response surface of the maximum total deformation at the first mode due to the static pressure of the fluid being accelerated by the 10-blade impeller.

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